A Daily Index of Economic and Political Uncertainty for Peru from X Discourse

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This version: October 22, 2026

Abstract

We construct a daily index of economic and political uncertainty for Peru from messages posted on X (formerly Twitter) by influential accounts between January 2018 and January 2023. A large language model, DeepSeek-V3, classifies each message along economic, political, and uncertainty dimensions. The headline index, EPU_t, is the volume-weighted fortnightly share of messages that meet all three criteria; a companion index, PU_t, drops the economic requirement and measures political uncertainty. EPU_t co-moves moderately with the Chicago Board Options Exchange Volatility Index (VIX) and more weakly with the conditional variance of Lima stock-market returns, and its association with the VIX is stronger during documented crises. Daily changes in the index and in the VIX are essentially uncorrelated, so the association reflects differences across episodes rather than day-to-day tracking. The two text indices identify different episodes: EPU_t discriminates economic crises well, whereas PU_t captures domestic political ruptures that market benchmarks largely miss. As an example of use, a bivariate vector autoregression associates increases in PU_t with a larger same-day volume of central-bank foreign-exchange intervention that is reversed the next day. The indices complement financial-market measures for monitoring uncertainty in Peru.

Keywords: Economic Political Uncertainty, X (formerly Twitter), Large Language Models, social media, DeepSeek-V3, GARCH, central bank intervention, Peru.

JEL Classification: C32, C43, C55, C82, D80, E44, G15.

Declaration on the Use of AI Tools: The authors used Claude Opus 4.8, 5 (Anthropic) for drafting and editing the LaTeX source, and Refine Ink for manuscript feedback. The authors reviewed all AI-assisted content and take full responsibility for it; all errors are their own.

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1 Introduction

Uncertainty can delay investment and hiring and alter financial risk premia (Bloom, 2009, 2014), particularly in emerging economies (Frankel, 2010; Carrière-Swallow and Céspedes, 2013). Public narratives and media coverage also shape economic decisions (Shiller, 2017; Veldkamp, 2006). Evidence from the financial press, internet searches, and social media motivates using text to measure these conditions at high frequency (Tetlock, 2007; García, 2013; Da et al., 2015; Baker et al., 2021).

The economic policy uncertainty index of Baker et al. (2016) counts newspaper articles mentioning the economy, policy, and uncertainty. More recent work uses X (formerly Twitter): Becerra and Sagner (2023), for example, track Chilean media outlets. We instead construct a daily index from influential individual accounts in Peruvian politics, economics, journalism, and business. This source captures discourse beyond institutional news feeds.

Our classification follows advances in text-based economic measurement (Gentzkow et al., 2019). Dictionary methods (Loughran and McDonald, 2011) provide transparent rules, while contextual models can distinguish meanings that keyword counts miss (Hassan et al., 2019; Devlin et al., 2019; Brown et al., 2020; Korinek, 2023). We use DeepSeek-V3 to label each tweet as economic (E), political (P), and uncertain (U). The headline economic and political uncertainty index, EPU_t, uses the tweets in the intersection $E \cap P \cap U$, which is conceptually distinct from newspaper-based *policy* uncertainty. The companion political-uncertainty index, PU_t, uses $P \cap U$ and therefore includes messages without economic content.

The sample spans January 2018–January 2023, a period containing six presidents, a dissolved Congress, a contested election, an attempted self-coup, the pandemic, and sustained protests. These events allow us to assess economic and political uncertainty within one country. Earlier platform coverage was sparse, while retrieval and classification costs limit extending the sample.

The paper contributes the daily index family and three empirical assessments. First, we compare EPU_t with three market benchmarks: the Chicago Board Options Exchange Volatility Index (VIX), the squared daily log return of the Bolsa de Valores de Lima (BVL) index, and the conditional variance of that return from a GARCH$(1,1)$ model (Bollerslev, 1986). Second, we assess whether these associations change during crises dated from historical events (Romer and Romer, 2010; Caldara and Iacoviello, 2022), and whether the indices discriminate crisis days from calm days. Third, as an example of how the indices can be used, a bivariate vector autoregression relates PU_t to central-bank foreign-exchange intervention. Its same-day response depends on the recursive identification and does not establish causality.

Section 2 describes the data, Section 3 constructs the indices, and Section 4 presents the methods. Section 5 dates the crisis windows and reports the results, and Section 6 concludes. Appendix A collects the econometric background, and Appendices B and C contain the tables and figures.

2 Data

2.1 Tweet Collection

We download tweets from 97 accounts representing the main agents in Peruvian political and economic discourse, spanning 23 January 2018 to 31 January 2023, across five categories: journalists (24), politicians (42), economists (16), political analysts (13), and private-sector representatives (2). Accounts are selected for public visibility (follower count, media citations, and a media presence that is common knowledge in Peruvian society), sustained activity across the window, and coverage of the main institutional actors (Congress, the executive, the central bank, major media outlets, among others). Appendix B, Table 1 lists the full set. This elite focus limits representativeness

relative to a full-population sample but follows Baker et al. (2016) in targeting "informed agents," whose discourse is more likely to reflect genuine policy uncertainty than noise.

The data are obtained via the X (formerly Twitter) application programming interface (API), which yields one comma-separated (CSV) file per account. Appendix C, Figure 1 summarizes the full processing pipeline in four stages. First, in the ingestion stage, all available tweets of each account within the sample window are pulled from the API into per-account raw files, the 101 accounts of the initial download. Second, an Extract–Transform–Load step merges, deduplicates, and cleans these files; it is here that the four institutional accounts (corporate and academic broadcast feeds) are dropped, reducing the 101 ingested accounts to the 97 individual accounts retained for analysis and leaving a consolidated corpus of 213 249 tweets. Third, every surviving tweet is sent to the DeepSeek-V3 API and labelled along the economic, political, and uncertainty dimensions, yielding a tweet-level uncertainty dataset. Fourth, this labelled dataset feeds the econometric analysis. We keep all items (original tweets, retweets, replies, and quoted posts), so that each retained item is a single tweet-level observation; the index therefore measures the intensity of discourse.

2.2 Semantic Classification via a Large Language Model (LLM)

The original Baker et al. (2016) approach classifies a newspaper article as policy-uncertainty-related if it contains at least one keyword from each of three pre-specified dictionaries: an "economic" list D_E (e.g., "economy", "inflation"), a "policy" list D_P (e.g., "Congress", "legislation"), and an "uncertainty" list D_U (e.g., "uncertain", "risk").

Formally, let x_a denote the raw text of article a (a string of tokens) and let $\text{tokens}(x_a)$ be the multi-set of words it contains. Let $w_E \in D_E$ denote any word belonging to the economic dictionary, and define $w_P \in D_P$, $w_U \in D_U$ analogously. The Baker et al. (2016) classifier assigns a binary label $z_a^{\text{BBD}} \in \{0, 1\}$ to article a:

$$z_a^{\text{BBD}} = \mathbf{1}\{\exists\, w_E \in D_E,\, w_P \in D_P,\, w_U \in D_U : w_E, w_P, w_U \in \text{tokens}(x_a)\},$$

so $z_a^{\text{BBD}} = 1$ when the article contains at least one representative term of each of the three dictionaries simultaneously. This deterministic rule misses synonyms, idiomatic expressions, irony, and context-dependent meaning.

We replace this rule with a semantic classifier based on DeepSeek-V3. Let x_{ijt} denote the text of the j-th tweet posted by user i on day t, where $i = 1, \ldots, 97$ indexes the X accounts, t indexes calendar days in the sample window, and $j = 1, \ldots, n_{it}$ indexes that user's tweets on day t. For each tweet, the classifier returns a triple of binary labels:

$$z_{ijt} = (E_{ijt},\, P_{ijt},\, U_{ijt}) \in \{0, 1\}^3,$$

where $E_{ijt} = 1$ indicates that tweet (i, j, t) has economic content, $P_{ijt} = 1$ indicates political content, and $U_{ijt} = 1$ indicates the expression of uncertainty.

The three dimensions are produced independently, by three separate expert-level prompts evaluated on the same tweet, each returning a single binary token. Appendix C, Figure 2 shows the anatomy of one such prompt. It is assembled from six blocks: a role description that casts the model as a panel of domain experts (economist, financial analyst, and so on); a task description stating the binary question for that dimension; solution guidance and a domain-knowledge block that delimit, with Peruvian-context vocabulary, what is to count as economic, political, or uncertainty-laden content; an exception-handling block instructing the model to read tweets in any language and to answer without explanation; and an output-formatting block that constrains the reply to a single 0 or 1. Appendix C, Figure 3 shows how the three prompts combine: the same tweet is passed in

parallel to the economic, policy, and uncertainty prompts, and the large language model returns the triple (E, P, U). For the tweet illustrated there, the model returns $(0, 1, 1)$ (no explicit economic content, but political content and an expression of uncertainty) so the tweet is counted in the political-uncertainty cell rather than the strict EPU cell.[1]

We adopt DeepSeek-V3 for three practical reasons: strong multilingual instruction-following performance, including Spanish; an open-weights release that makes the classification step replicable; and an inference cost low enough to label the full corpus along the three dimensions (approximately 160 United States dollars (USD)) a small fraction of the cost of comparable proprietary models at the time of classification.

Appendix B, Table 2 reports the eight possible (E, P, U) label combinations with their frequencies, and Appendix B, Table 3 summarizes the cleaned sample. Of the 213 249 tweets posted by the 97 accounts between January 2018 and January 2023, about 38% are unrelated to economics, politics, or uncertainty, while close to half carry political content. Uncertainty in Peruvian elite discourse is overwhelmingly political rather than economic: the political-uncertainty cell $(0, 1, 1)$ alone contains 59 978 tweets, against only 1 318 tagged as purely economic uncertainty $(1, 0, 1)$. The strict label $(1, 1, 1)$, which requires the simultaneous presence of all three dimensions, is comparatively rare, 16 451 tweets, 7.7% of the corpus, and this sparsity is precisely what motivates the complementary political-uncertainty index built from the $P \cap U$ cells.

The 16 451 strict tweets occur on 1 462 distinct calendar days, so the mean on days with a positive count is $16\,451/1\,462 = 11.25$ tweets. Daily tweet volume, N_t, the number of classified tweets posted on calendar day t, is strongly right-skewed: its mean (116.21) far exceeds its median (56), and the maximum is 1 016 tweets in a single day, reflecting sharp bursts of activity around political–economic events. The index is thus well populated yet concentrated on event-driven spikes, consistent with the state-dependent behavior documented below.

3 The Index

We build the headline daily index and its political companion from the same classified tweets. Because every step of the construction is a choice (what to count, how to normalize it, how much to smooth it, and how to scale it) we make those choices, and the reasons for them, explicit. Each is defended below on statistical and substantive grounds, and its effect is shown on the data.

For each calendar day t, let $N_t = \sum_{i=1}^{97} n_{it}$ be the number of classified tweets posted that day and let Y_t be the number of those tweets that are economic, political, and uncertain at the same time:

$$Y_t \equiv Y_t^{E \cap P \cap U} = \sum_{i=1}^{97} \sum_{j=1}^{n_{it}} \mathbf{1}\{E_{ijt} = 1,\ P_{ijt} = 1,\ U_{ijt} = 1\}, \tag{1}$$

where $E_{ijt}, P_{ijt}, U_{ijt} \in \{0, 1\}$ are the economic, political, and uncertainty labels of the j-th tweet posted by account i on day t, and n_{it} is the number of tweets that account posted that day. Y_t is a count of tweets, not an index. The two broader cells that drop one requirement are counted over the same tweets:

$$Y_t^{E \cap U} = \sum_{i=1}^{97} \sum_{j=1}^{n_{it}} \mathbf{1}\{E_{ijt} = 1,\ U_{ijt} = 1\}, \qquad Y_t^{P \cap U} = \sum_{i=1}^{97} \sum_{j=1}^{n_{it}} \mathbf{1}\{P_{ijt} = 1,\ U_{ijt} = 1\}.$$

Appendix C, Figure 4 plots these three daily counts. Two things stand out. First, the strict count Y_t (bottom panel) and the economic-uncertainty count $Y_t^{E \cap U}$ (top panel) are almost indistinguishable

[1] The labelling was monitored manually: two of the authors (Gallardo and Loaiza) inspected samples of classified tweets against their returned labels throughout the run.

in level and shape: over the sample the strict cell holds 16 451 tweets and $E \cap U$ holds 17 769, and 93 percent of the latter are also political, so the two are nearly the same object. The political-uncertainty count $Y_t^{P \cap U}$ (middle panel) is an order of magnitude larger (76 429 tweets, hundreds on a busy day against tens for the other two) and moves on its own schedule, spiking on episodes of purely political turbulence that carry no economic content. This is why the headline index counts only the strict intersection: relaxing the economic requirement would swap in the much larger, differently timed political signal. Second, all three counts are spiky and drift upward through the sample as these accounts post more, so a raw count confounds uncertainty with sheer activity.

We therefore measure the share of the day's discourse that is uncertain, normalizing Y_t by N_t:

$$s_t \;=\; \frac{Y_t}{N_t} \qquad (0 \le s_t \le 1), \tag{2}$$

and likewise for the two broader cells. Appendix C, Figure 5 plots these shares in the same order as Appendix C, Figure 4. The mechanical volume component is gone (the upward drift of the counts flattens) but the daily series is now noisy and ill-behaved on thin days: when only a handful of tweets are posted the denominator N_t is tiny, so a single tweet moves s_t by a large amount and the share can jump close to 1. Daily volume averages 116 tweets but its median is only 56, so thin days are common, and the early, low-volume part of the sample (2018) is correspondingly jagged.

To control that small-sample noise we pool counts over a trailing window of W calendar days, from $t-W+1$ to t, and take the ratio of the summed counts (a volume-weighted rate) rather than the average of the daily shares:

$$\rho_t \;=\; \frac{\sum_{s=t-W+1}^{t} Y_s}{\sum_{s=t-W+1}^{t} N_s}. \tag{3}$$

Pooling counts is not the same as averaging the daily shares of Eq. (2). The denominator is the number of classified tweets in the window, $\mathcal{N}_t = \sum_{s=t-W+1}^{t} N_s$, rather than the volume of day t alone. Rearranging Eq. (3) expresses ρ_t as a weighted mean of the daily shares:

$$\rho_t \;=\; \sum_{s=t-W+1}^{t} \frac{N_s}{\mathcal{N}_t}\, s_s \;=\; \sum_{s=t-W+1}^{t} w_s\, s_s, \qquad w_s = \frac{N_s}{\mathcal{N}_t}, \qquad \sum_{s=t-W+1}^{t} w_s = 1, \tag{4}$$

so each daily share s_s is weighted by that day's volume N_s. A high-volume day contributes in proportion to how much it actually says, and a low-volume day cannot dominate the window. For $t < W$ the sums in Eq. (3) run over the available days $s = 1, \ldots, t$, so the first $W-1$ dates use an expanding window.

We set $W = 14$ calendar days. The choice trades precision against responsiveness, and the data do not identify an interior optimum. Precision can be gauged by treating the $\mathcal{N}_t$ tweets in the window as independent draws with success probability ρ_t. Under this binomial approximation, the standard error of ρ_t is:

$$\mathrm{se}_t \;=\; \left\{ \frac{\rho_t\,(1-\rho_t)}{\mathcal{N}_t} \right\}^{1/2}. \tag{5}$$

Longer windows raise $\mathcal{N}_t$ and therefore lower se_t. They also delay the index, because a trailing window of W days is centred $(W-1)/2$ days before t (its phase lag), and they flatten the short-lived episodes the index is meant to capture. The two extremes are degenerate: minimizing noise alone drives W toward the sample length and produces a constant series, whereas minimizing the lag returns the noisy daily share of Eq. (2). A fortnight lies between the daily frequency at which the text arrives and the monthly frequency of most activity indicators.

Appendix C, Figure 6 quantifies this trade-off for nine window lengths: $W = 3$, 5, 7, 10, 14, 21, 30, 45, and 60 days. Each point is one window, placed on the horizontal axis at its phase lag.

The black curve (left axis) is the noise ratio: the sample average of se_t divided by the standard deviation of ρ_t. It compares the typical sampling error with the variation of the rate over time and is a scale-free summary, not a formal variance share. The red curve (right axis) is the highest value of the rate inside the COVID-19 emergency window C3 (15 March–30 June 2020, one of the eight crisis windows dated in Section 5.1), in multiples of the rate's sample mean; it measures how much of the largest episode a window retains. Both statistics use only dates with a complete window, $t \geq W$, so that the expanding start of Eq. (3) does not favour long windows. The noise ratio falls monotonically, from 0.48 at $W = 3$ to 0.33 at $W = 14$, 0.25 at $W = 30$, and 0.19 at $W = 60$. The retained peak also falls, from 3.54 times the mean at $W = 3$ to 2.14 at $W = 60$, but not uniformly: it drops steeply up to $W = 10$, stays at 2.72 between $W = 10$ and $W = 14$, and declines again for longer windows. The baseline therefore sits at the end of a stretch in which lengthening the window still buys precision (the noise ratio falls by about a tenth between $W = 10$ and $W = 14$) without losing amplitude in the largest episode, at the cost of a 6.5-day phase lag. Longer windows keep reducing noise, but each further reduction costs peak amplitude and adds lag. The figure does not identify an optimal window; it shows that $W = 14$ is not an extreme point of the trade-off.

The main episodes do not depend on the fortnightly window. For each alternative W we rebuild the index with Eqs. (3) and (6) below and compute its Pearson correlation with the baseline $W = 14$ index over all 1 835 calendar days. The correlations are 0.88 for $W = 7$, 0.94 for $W = 21$, and 0.89 for $W = 30$. Among these three alternatives, the 21-day index correlates most with the baseline because it is the closest in construction, not because it is more accurate. Two features of the construction account for the ranking. First, the windows are nested. Every day of the baseline window also enters the 21-day window and makes up two thirds of it (14/21), whereas the 7-day window covers half of the baseline window (7/14) and the baseline covers less than half of the 30-day window (14/30); the more days two windows share, the more tweets their rates have in common. Second, the windows smooth differently. The 7-day index carries more sampling noise (a noise ratio of 0.41 against 0.33 in Appendix C, Figure 6), which lowers its correlation with any smoother series, whereas the 30-day index averages over a longer stretch and is centred eight days earlier than the baseline. Timing explains little of the gap: shifting each alternative index to its best alignment with the baseline raises the correlations only to 0.89, 0.95, and 0.90. The overlap of the windows and their different degree of smoothing are therefore the more plausible explanations, although these calculations do not separate their contributions exactly. In either case, a higher correlation with the baseline indicates a similar construction; it is not evidence of greater precision or validity.

Following Baker et al. (2016), we rescale the rate by its own sample mean so that the series reads in interpretable units:

$$\mathrm{EPU}_t \;=\; 100 \cdot \frac{\rho_t}{\bar{\rho}}, \qquad \bar{\rho} \;=\; \frac{1}{T}\sum_{t=1}^{T} \rho_t, \tag{6}$$

where $T = 1\,835$ is the number of calendar days. A value of 100 denotes the sample-average rate, and 200 denotes twice that average share of messages. The mean rolling rate is $\bar{\rho} \approx 0.073$, so on average about 7.3% of the pooled discourse jointly meets the economic, political, and uncertainty criteria. Rescaling leaves the time pattern unchanged.

The same pooling and scaling steps applied to the political-uncertainty cell give the companion index. Let $\rho_t^{P\cap U} = \sum_{s=t-W+1}^{t} Y_s^{P\cap U} / \sum_{s=t-W+1}^{t} N_s$ be the volume-weighted fortnightly rate of that

cell, computed exactly as in Eq. (3) but with $Y_s^{P\cap U}$ in the numerator. The political-uncertainty index is that rate on the same base-100 scale:

$$\mathrm{PU}_t \;=\; 100 \cdot \frac{\rho_t^{P\cap U}}{\bar{\rho}^{P\cap U}}, \qquad \bar{\rho}^{P\cap U} \;=\; \frac{1}{T}\sum_{t=1}^{T} \rho_t^{P\cap U}, \tag{7}$$

so that PU_t and EPU_t are read in the same units, each equal to 100 on an average day of its own cell. PU_t does not require economic content. We carry it alongside EPU_t because political crises can generate uncertain messages without an explicit economic frame. We do not build a separate index from the $E \cap U$ cell, which nearly coincides with the strict cell.

Appendix C, Figure 7 plots EPU_t against its two simpler forms, the raw count $100\, Y_t/\bar{Y}$ and the daily share $100\, s_t/\bar{s}$, each rescaled by its own mean so that all three average 100 by construction. The mean therefore cannot separate them; the median can, and the further it falls below 100 the more the average is carried by a few extreme days. The raw count has a median of 44.6 and a maximum of 915, because a handful of high-volume days carry the average; the daily share, a median of 86.3 and a maximum of 1 343, because a thin denominator lets one tweet dominate. EPU_t has a median of 96.4 and a maximum of 272: its typical day sits close to the base, and its extremes are episodes rather than artifacts of volume or of a small denominator.

4 Methodology

We assess the indices through static and rolling correlations with market benchmarks, comparisons between crisis and calm periods, and their ability to discriminate crisis days from calm days. We then illustrate one use of the indices with central-bank foreign-exchange intervention. These exercises evaluate association and predictive content; perfect agreement with a benchmark is not expected because the measures capture different sources of uncertainty.

4.1 Benchmark Indicators

We use three market benchmarks, which differ in nature, units, and horizon (Appendix B, Table 4). The VIX is the Chicago Board Options Exchange index of the volatility of the United States S&P 500 index implied by option prices. It is a standard deviation, quoted in annualized percentage points, and it refers to the next 30 calendar days. We use the daily CBOE close, VIX_t.

For the Bolsa de Valores de Lima (BVL), let P_t be the closing level of the S&P/BVL index on trading date t, and let $r_t = \log(P_t/P_{t-1})$ be its log return in decimal units, where $t-1$ is the previous date of the trading-day panel described below. The second benchmark is the squared return r_t^2. It is a noisy, one-day measure of return variance, in squared decimal units, and it looks only at the current day: its dependence vanishes within a day (Appendix B, Table 5). The third benchmark is the conditional variance $\hat{\sigma}^2_{\mathrm{BVL},t}$ of r_t from the GARCH$(1,1)$ model of Section 4.4. It is a model-based forecast of the variance of r_t, made with returns up to the previous trading date, in the same units as r_t^2 but much smoother.

The VIX therefore measures volatility (a standard deviation), whereas r_t^2 and $\hat{\sigma}^2_{\mathrm{BVL},t}$ measure variance. Pearson correlations are invariant to changes of units and origin but not to squaring, so a correlation with a volatility measure is not identical to the correlation with the corresponding variance. Rank-based statistics, the Spearman correlation and the area under the receiver operating characteristic curve of Section 5.3, are invariant to such monotone transformations. All three benchmarks measure financial-market risk, whereas EPU_t measures classified discourse about Peru.

Financial comparisons use common trading dates. The supplied BVL and PEN/USD (Peruvian Sol per US dollar) files contain linearly interpolated prices on non-trading dates, which would suppress measured volatility. We identify these values from vanishing second differences and compare the flagged dates with the holiday calendar. The procedure also flags an interpolated BVL stretch from 25 January to 2 March 2021, which we exclude. For the VIX we use the CBOE close directly.[2]

The intersection contains 1 190 quoted dates. Returns are computed between consecutive dates of this panel, so a return that follows a holiday in Lima, in New York, or in the foreign-exchange market spans more than one session. One return, on 3 March 2021, spans the excluded stretch (40 calendar days) and is discarded; no other return spans more than six calendar days. The comparison panel therefore has $T = 1\,188$ dates with a valid daily return. Using the eight crisis windows dated in Section 5.1, it contains 234 crisis dates (84 in economic and 150 in political windows) and 954 calm dates. The text indices remain defined on all 1 835 calendar days and are sampled on these dates without interpolation.

4.2 Static Co-Movement

Let b_t denote the benchmark compared with EPU_t, namely VIX_t, r_t^2, or $\hat{\sigma}^2_{\mathrm{BVL},t}$, on the T dates of the comparison panel. The Pearson correlation measures linear association:

$$\rho \;=\; \frac{\sum_{t=1}^{T}(\mathrm{EPU}_t - \overline{\mathrm{EPU}})(b_t - \bar{b})}{\sqrt{\sum_{t=1}^{T}(\mathrm{EPU}_t - \overline{\mathrm{EPU}})^2}\,\sqrt{\sum_{t=1}^{T}(b_t - \bar{b})^2}}, \tag{8}$$

where $\overline{\mathrm{EPU}}$ and $\bar{b}$ are the means over the T dates. With ranks $R_t = \mathrm{rank}(\mathrm{EPU}_t)$ and $Q_t = \mathrm{rank}(b_t)$, the Spearman correlation is the Pearson correlation of the ranks:

$$\rho_S \;=\; \frac{\sum_{t=1}^{T}(R_t - \bar{R})(Q_t - \bar{Q})}{\sqrt{\sum_{t=1}^{T}(R_t - \bar{R})^2}\,\sqrt{\sum_{t=1}^{T}(Q_t - \bar{Q})^2}}. \tag{9}$$

We use the Spearman correlation as a descriptive robustness check, because it is unaffected by monotone transformations such as squaring, and we conduct inference on the Pearson correlation.

Appendix B, Table 5 reports persistence diagnostics for EPU_t, VIX_t, r_t^2, and $\hat{\sigma}^2_{\mathrm{BVL},t}$. The augmented Dickey–Fuller (ADF) test (Dickey and Fuller, 1979; Said and Dickey, 1984) has a unit-root null. We include a constant and choose the number of lagged differences by the Schwarz (Bayesian) information criterion (BIC), up to 23 lags (Appendix A.1). All four series reject the unit root at the 1% level. The table also reports the first-order autocorrelation $\hat{\rho}_1$ and the implied half-life $\ln(1/2)/\ln\hat{\rho}_1$ in trading days. EPU_t, VIX_t, and $\hat{\sigma}^2_{\mathrm{BVL},t}$ are nevertheless highly persistent, with half-lives of 17 to 22 trading days, whereas r_t^2 loses its dependence within a day. We therefore interpret correlations in levels cautiously and also examine first differences.

Serial dependence matters because consecutive values of EPU_t share most of their fourteen days of tweets (thirteen on consecutive calendar days and at least eight on consecutive dates of the comparison panel) and the financial series are persistent. We estimate standard errors of the Pearson correlation with the heteroskedasticity- and autocorrelation-consistent (HAC) estimator of Newey and West (1987), using the weights of Bartlett (1950). Appendix A.2 derives the correlation's influence function, its long-run variance, and the adaptive bandwidth rule L^A. We report that bandwidth alongside fixed bandwidths $L \in \{7, 30, 60, 100\}$ and, for comparison, the classical standard error $(1 - \hat{\rho}^2)/\sqrt{T}$, which ignores serial dependence.

[2]The supplied VIX file equals the CBOE series multiplied by the PEN/USD exchange rate; we use the original CBOE series.

Correlations are estimated on the full panel and separately on crisis and calm dates. The windows C1–C8 of Section 5.1 are dated from historical events, independently of market movements. Pooled crisis correlations can reflect differences across episodes and need not imply co-movement within each episode. To assess sensitivity to the dating, we shift all windows jointly by -5 and by $+5$ calendar days and recompute the crisis and calm correlations.

4.3 Time-Varying Co-Movement

To see how the association changes over time, we also compute the Pearson correlation over a trailing window of $m = 90$ consecutive dates of the comparison panel, about four and a half calendar months. For the date t that closes the window, the rolling correlation is:

$$\widehat{\mathrm{corr}}_t^{(m)} = \frac{\sum_{s=t-m+1}^{t}(\mathrm{EPU}_s - \overline{\mathrm{EPU}}_t)(b_s - \bar{b}_t)}{\sqrt{\sum_{s=t-m+1}^{t}(\mathrm{EPU}_s - \overline{\mathrm{EPU}}_t)^2}\,\sqrt{\sum_{s=t-m+1}^{t}(b_s - \bar{b}_t)^2}}. \tag{10}$$

Here b_s is the benchmark on date s, and $\bar{b}_t = m^{-1}\sum_{s=t-m+1}^{t} b_s$ is its average over the same m dates that end at t; $\overline{\mathrm{EPU}}_t$ is the corresponding average of the index. The sums run over rows of the trading-day panel, not over calendar days, so each window contains exactly 90 valid trading dates, and the first value is available once 90 dates have accumulated. Because $\bar{b}_t$ and $\overline{\mathrm{EPU}}_t$ are recomputed for every t, the rolling correlation measures co-movement around the local levels of the two series over the window rather than around their full-sample means: a change in average levels from one period to the next does not by itself produce a positive value. Section 5.3 applies the formula with $b_t = \mathrm{VIX}_t$ and $b_t = \hat{\sigma}^2_{\mathrm{BVL},t}$, and to first differences of EPU_t and VIX_t.

The window balances sampling noise against timing resolution. Persistence reduces the effective sample size, so the curves are descriptive. Each window can combine crisis and calm dates; even the longest crisis windows contain fewer than 90 trading dates. Short episodes are therefore assessed mainly through the crisis–calm comparisons of Section 4.2.

4.4 Conditional Variance of BVL Returns

We model the conditional variance of BVL returns with a GARCH$(1,1)$ (Bollerslev, 1986), which extends the ARCH model of Engle (1982) by adding the lagged conditional variance. This is the only GARCH model in the paper: it is estimated on the return r_t, not on r_t^2, and not on the text indices or the VIX. Let $\mathcal{F}_{t-1}$ denote the information available at the close of the previous trading date. The return equation is:

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t\, z_t, \qquad z_t \sim \mathrm{N}(0,1) \text{ i.i.d.}, \tag{11}$$

where μ is the mean return and $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$ is the conditional variance. The conditional variance follows the recursion:

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \qquad \omega > 0, \quad \alpha \geq 0, \quad \beta \geq 0, \quad \alpha + \beta < 1. \tag{12}$$

The coefficient α measures the response of the variance to the previous squared innovation, while β carries past variance forward; their sum $\alpha + \beta$ measures persistence. We maximize the Gaussian quasi-likelihood (Bollerslev and Wooldridge, 1992) and, as a robustness check of the same model, also use standardized Student-t innovations. For numerical stability the model is estimated on $100\,r_t$, in percentage points; the conditional variance is then converted back to squared decimal units, the units of r_t^2. Appendix A.3 gives the likelihood and the variance properties behind the statistics of Appendix B, Table 6.

With the fitted innovations $\hat{\varepsilon}_t = r_t - \hat{\mu}$, the filtered conditional variance is:

$$\hat{\sigma}^2_{\mathrm{BVL},t} \;=\; \hat{\omega} + \hat{\alpha}\,\hat{\varepsilon}^2_{t-1} + \hat{\beta}\,\hat{\sigma}^2_{\mathrm{BVL},t-1}, \qquad t = 2, \ldots, T. \tag{13}$$

Its square root, $\hat{\sigma}_{\mathrm{BVL},t}$, is the conditional standard deviation, or volatility. The comparisons use the variance $\hat{\sigma}^2_{\mathrm{BVL},t}$, the quantity that r_t^2 measures with noise. We assess remaining volatility clustering with Ljung–Box tests (Ljung and Box, 1978) on the squared standardized residuals $\hat{z}_t^2 = \hat{\varepsilon}_t^2/\hat{\sigma}^2_{\mathrm{BVL},t}$. Appendix B, Table 6 reports parameter estimates, persistence, log-likelihoods, and residual diagnostics. Persistence close to one (Engle and Bollerslev, 1986) makes implied half-lives particularly sensitive to estimation error.

The main comparison uses the levels of EPU_t and VIX_t and the conditional variance $\hat{\sigma}^2_{\mathrm{BVL},t}$. We standardize them to zero mean and unit variance for display and report Pearson correlations for the full panel, crisis dates, and calm dates; standardization leaves these correlations unchanged.

4.5 An Example Specification: Uncertainty and Foreign-Exchange Intervention

As an example of how the indices can be used, we relate them to the foreign-exchange intervention of the Banco Central de Reserva del Perú (BCRP), which buys and sells US dollars to limit fluctuations in the value of the Sol. We estimate a bivariate vector autoregression (VAR) in $\mathbf{z}_t = (X_t, y_t)'$:

$$\mathbf{z}_t = \mathbf{c} + \sum_{j=1}^{p} A_j \mathbf{z}_{t-j} + \mathbf{u}_t, \qquad \mathrm{Var}(\mathbf{u}_t) = \Sigma_u. \tag{14}$$

Here X_t is PU_t or, for comparison, EPU_t; $\mathbf{c}$ is a vector of intercepts, $A_1, \ldots, A_p$ are 2×2 coefficient matrices, and $\mathbf{u}_t$ are reduced-form innovations with covariance matrix Σ_u. The outcome y_t is either the absolute value of the BCRP's daily foreign-exchange operations including swaps, $|\mathrm{INT}_t|$, or the absolute value of its net spot purchases, $|\mathrm{COM}_t|$, in millions of US dollars. Taking absolute values measures the volume of intervention whether the BCRP buys or sells dollars, so the exercise concerns how much the central bank operates, not the direction of its operations.

The BIC (Schwarz, 1978) selects the lag order p separately by sample: from one to six lags for the full sample and from one to three for the crisis blocks. The crisis blocks stack the windows of Section 5.1, and no lag crosses a window boundary; C4 and C7 are excluded because they have at most eight trading dates. Full-sample intervals are asymptotic; crisis intervals use 500 residual-bootstrap replications within the window blocks. A Cholesky factorization of Σ_u orders uncertainty first, allowing intervention to respond within the day but ruling out the reverse contemporaneous effect. Same-day responses depend on this assumption; Granger tests separately assess whether past uncertainty predicts intervention.

4.6 What the Index May Add and Its Limitations

The tools above speak to convergent validity: agreement with the VIX and with the variance of BVL returns would support the claim that the index measures uncertainty. Perfect agreement is neither expected nor desirable. EPU_t captures economic and political uncertainty in discourse concerning Peru; PU_t also includes political uncertainty without economic content. Both are built from the messages of Peruvian politicians, journalists, economists, and analysts, so they register events (presidential vacancy votes, cabinet collapses, contested elections) that a United States equity-option index reflects weakly and that domestic returns may reflect only with delay. We document where the indices move while the benchmarks do not, and the reverse.

Appendix B, Table 4 sets these differences out. Beyond the domestic and political focus, the measures differ in orientation and anchoring: the index reads uncertainty as expressed in discourse,

the VIX as the market's expectation of future volatility, and r_t^2 and $\hat{\sigma}^2_{\mathrm{BVL},t}$ as summaries of past returns, with only the benchmarks tied to a traded price. This is why we expect moderate rather than perfect co-movement, and why the index is not redundant with any benchmark.

The comparisons separate crisis days from calm ones, on the expectation that a measure of uncertainty tracks the benchmarks most closely when uncertainty is high. The split rests on a calendar fixed in advance, dated from the historical record rather than from the benchmarks it evaluates. Section 5.1 sets out these windows for Peru over 2018–2023, including the pandemic.

5 Empirical Results

We apply the tools of Section 4 to EPU_t and PU_t of Eqs. (6) and (7). The indices are defined on all 1 835 calendar days; every statistic involving a market series is computed on the comparison panel of Section 4.1. Each statistic is invariant to the base-100 convention.

5.1 Crisis Periods in Peru, 2018–2023

The crisis–calm comparisons require an explicit list of crisis windows. We date them from the documented political, institutional, and macroeconomic record, following the narrative approach of Romer and Romer (2010) and event-based risk indices (Caldara and Iacoviello, 2022). An episode qualifies when it combines (i) a datable triggering event, (ii) a rupture in an institution (the presidency, Congress, or the electoral authority) or a nationwide economic shock, and (iii) sustained attention rather than a one-day headline. In five years Peru had six presidents, a dissolved Congress, a contested election, an attempted self-coup, and a pandemic. Appendix B, Table 7 lists the eight windows and classifies each as economic or political by the origin of the episode; the classification uses no information from any series.

C1. Fall of Kuczynski (March 2018). Pedro Pablo Kuczynski resigned the presidency on 21 March 2018 after the release of vote-buying recordings, and Martín Vizcarra was sworn in on 23 March. We date it 12 March–6 April 2018.

C2. Dissolution of Congress (September–October 2019). On 30 September 2019 Vizcarra dissolved Congress, which contested the act and swore in Vice President Mercedes Aráoz, who resigned within a day; the clash briefly left two claimants to the executive. We date it 27 September–15 October 2019.

C3. COVID-19 emergency (March–June 2020). The state of emergency and lockdown were declared on 15 March 2020, the sharpest output contraction in the sample and coincident with the global risk-off shock. It is the one window macroeconomic and global rather than domestic-political. We date the acute phase 15 March–30 June 2020, ending at the first major easing.

C4. Vacancy of Vizcarra and Merino interregnum (November 2020). Congress vacated Vizcarra on 9 November 2020; Manuel Merino took office on 10 November and resigned on 15 November after protests in which two demonstrators were killed; Francisco Sagasti was sworn in on 17 November. Three presidents in one week. We date it 9–18 November 2020.

C5. Contested general election (April–July 2021). The first round (11 April 2021) and the runoff (6 June) between Pedro Castillo and Keiko Fujimori were followed by weeks of annulment disputes before the electoral authorities; Castillo was proclaimed on 19 July and inaugurated on 28 July. We date it 8 April–28 July 2021, from three days before the first round to the inauguration.

C6. Fuel-price protests and Lima stay-at-home order (March–April 2022). Rising fuel and fertilizer prices after the invasion of Ukraine triggered a nationwide transport strike. Near

midnight on 4 April 2022 the government issued Supreme Decree 034-2022-PCM, ordering a 22-hour mandatory stay-at-home period for Lima and Callao on 5 April; it was revoked that same afternoon amid open defiance. Economic in origin, institutional in expression, it is classified as economic. We date it 28 March–12 April 2022.

C7. Castillo self-coup and removal (December 2022). On 7 December 2022 Castillo announced the dissolution of Congress; within hours he was impeached, removed, and detained, and Dina Boluarte was sworn in. With C4, this is the sharpest institutional rupture of the sample. We date it 7–20 December 2022.

C8. Post-Castillo protests (December 2022–January 2023). Castillo's removal set off sustained, at times fatal, nationwide protests demanding new elections and Boluarte's resignation, past the sample end. We date it 20 December 2022–31 January 2023, right-censored by the sample end.

The eight windows cover 347 distinct calendar days out of 1 835 (18.9%), leaving 1 488 calm calendar days. On the comparison panel, the corresponding counts are 234 and 954 dates. Coverage is uneven by design: the pandemic (C3, 108 days) and the contested election (C5, 112 days) supply 63% of the window days, while the other six are 10-to-43-day shocks. Every year holds at least one window, so the split is not identified off a single episode. C7 and C8 are adjacent, sharing 20 December 2022; we keep them separate, as an elite institutional act differs from the popular mobilization that followed, and count the shared day once. The windows follow each event's timing: C7 begins on the self-coup date, whereas C1, C2, and C5 begin before their scheduled or anticipated triggers. We assess sensitivity by shifting their boundaries by five calendar days.

5.2 Behavior Across Crisis Windows

Appendix C, Figure 8 plots the two text indices and the three market benchmarks with the windows shaded. EPU_t averages 113 in crisis windows and 97 outside them, and reaches its sample maximum of 272 on 15 April 2020, inside C3. It rises in the economic windows but falls below its base in C4 (an average of 45) and C7 (93), where PU_t rises instead; PU_t averages 174 in C7. Political messages can satisfy $P \cap U$ without economic content, while the surge in total posting during political crises enlarges the denominator $\mathcal{N}_t$ and lowers the strict share. The market series reach their highest values around the onset of C3 in March 2020; the gap in early 2021 in their panels is the excluded BVL stretch.

Appendix C, Figure 9 isolates the political signal. It plots PU_t on all calendar days, with the political windows in amber and the economic windows in grey, and marks a threshold one standard deviation above the mean of PU_t (128); the lower panel indicates the days above it. The threshold is a simple descriptive rule, not an estimated cut-off. Excluding C5, 81% of political-window days lie above it, against 13% of calm days, so high readings of PU_t are concentrated in the political windows rather than spread across the sample. The share is highest in C7 (100%), C1 (92%), and C8 (84%), and the sample maximum of PU_t (180) falls on 20 December 2022, the day shared by C7 and C8. Only 4% of C3 days exceed the threshold, consistent with the pandemic being an economic rather than a political episode in the classified discourse. By contrast, no day of the contested election C5 does. PU_t averages 96 in C5, close to its base: the window, which runs from just before the first round to the inauguration, mixes a few acute weeks with months of ordinary political discourse. The figure shows in levels the pattern that the ranking statistics of Section 5.3 summarize: PU_t identifies most political windows clearly, the shortest one, C4, less so (40% of its days exceed the threshold), and the coarse dating of C5 dilutes the signal.

5.3 Similarity Analysis

Appendix B, Table 8 reports the static co-movement of EPU_t with VIX_t and r_t^2. Over the comparison panel, EPU_t and VIX_t correlate at 0.434 (Pearson) and 0.408 (Spearman); the two statistics are close, so the association is not the work of a few extreme days. The split by regime runs in the direction a measure of uncertainty should show: 0.648 on crisis dates against 0.261 on calm dates, with the Spearman values moving the same way (0.658 against 0.331). The association is more than twice as strong inside the eight crisis windows.

These correlations are computed on strongly dependent days, so the question is whether they remain distinguishable from zero once that dependence is priced into the standard errors. EPU_t is a trailing fourteen-day rate, so consecutive observations share most of their fourteen days of tweets, and the VIX is itself persistent. The dependence is visible in the score of Eq. (A.2), the daily term whose autocovariances determine the variance of $\hat{\rho}$: its first-order autocorrelation is 0.880. The classical standard error, which treats every day as fresh information, is 0.024 for the full-sample correlation of 0.434; the Newey–West estimator of Eq. (A.4), which sums those autocovariances, returns 0.085 (adaptive bandwidth $L^A = 10$), three and a half times larger. All three correlations survive the correction: 0.434 lies 5.1 of these standard errors from zero, the crisis value 0.648 lies 6.7 (HAC standard error 0.096), and the calm value 0.261 lies 3.5 (0.074). The crisis–calm ordering is also robust to the dating of the windows: shifting every start and end jointly by -5 or $+5$ days gives crisis correlations of 0.587 and 0.772 and calm correlations of 0.298 and 0.245, so the crisis value stays above the calm value under both shifts.

EPU_t shows no comparable relation with the squared return r_t^2: the Pearson correlation is 0.050 against a HAC standard error of 0.037, the crisis and calm values (0.010 and 0.047) are equally indistinguishable from zero, and the two shifts give -0.028 and 0.055 on crisis dates and 0.109 and 0.030 on calm dates. The HAC correction changes the standard error much less here (0.037 against a classical 0.029, compared with a factor of three and a half for the VIX), because r_t^2 has little persistence: the correction is large where persistence is high and small where it is not. The absence of a relation is a statement about frequency. r_t^2 reverts within a day, whereas EPU_t is a trailing fortnightly rate, so a single day of large price moves in Lima barely affects a two-week average. What the two might share is the smoother, persistent component of return variance, which $\hat{\sigma}^2_{\text{BVL},t}$ captures (Section 5.4).

The EPU_t–VIX association mainly reflects differences across episodes. Within individual windows, correlations range from -0.800 in C1 to 0.790 in C6 and equal 0.058 in the pandemic window C3; the short windows provide few observations. During C3, however, both series are far above their sample means (EPU_t averages 184 and VIX_t 39.7 on C3 dates, against 100 and 21.5 over the panel), which contributes much of the pooled crisis correlation of 0.648. The rolling correlation of Eq. (10) with the VIX (Appendix C, Figure 10) averages 0.052 and is negative in 42.5% of windows; it averages 0.247 in windows that end on crisis dates and 0.005 in windows that end on calm dates. The rolling correlation with $\hat{\sigma}^2_{\text{BVL},t}$ is less stable still: it averages -0.027 and is negative in 58% of windows.

Correlation in levels can arise because two persistent series rise and fall together across episodes, even if their daily news is unrelated. To separate these frequencies we also correlate first differences, $\Delta\text{EPU}_t = \text{EPU}_t - \text{EPU}_{t-1}$ and $\Delta\text{VIX}_t = \text{VIX}_t - \text{VIX}_{t-1}$, where $t-1$ is the previous date of the comparison panel, so a difference after a weekend or a holiday spans more than one calendar day. Differencing removes the slowly moving level that both series share during long episodes and asks a different question: on the days when one series moves, does the other move in the same direction? For EPU_t, a daily change reflects the tweets that enter the fourteen-day window and those that leave it, so ΔEPU_t is the index's daily news. Appendix C, Figure 11 contrasts the two frequencies.

Panel (a) overlays the rolling correlation of Eq. (10) in levels (solid) on the same statistic computed with first differences (dashed); panel (b) plots ΔEPU_t against ΔVIX_t. The full-sample correlation of the differences is -0.015, with a HAC standard error of 0.025. The rolling correlation of differences stays between -0.22 and 0.24 throughout, whereas the rolling correlation in levels exceeds 0.3 in absolute value in 39% of windows, and the scatter is a centred cloud with no discernible slope. The association between EPU_t and the VIX therefore lives at low frequency: the two rise and fall together across episodes, but the index does not echo the benchmark from one day to the next, and it cannot be read as a daily tracker of it. This is a statement about co-movement, not about causation in either direction; the VIX also reflects United States news that need not appear in Peruvian discourse.

A different question, and the one a user of the index would ask first, is whether it separates the days a historian would call a crisis from the days around them. To answer it we treat each series as a score and the crisis calendar as the label. For a threshold c, a date is classified as a crisis when the series exceeds c. The true positive rate (TPR) is the share of crisis dates above c, and the false positive rate (FPR) is the share of calm dates above c. The receiver operating characteristic (ROC) curve traces the TPR against the FPR as c moves from the highest to the lowest value of the series, from $(0, 0)$ to $(1, 1)$. The area under this curve (AUC) summarizes all thresholds at once: it equals the probability that a randomly drawn crisis date ranks above a randomly drawn calm date, with ties counted one half, and we estimate it with the Mann–Whitney statistic (Mann and Whitney, 1947). A series unrelated to the calendar has a ROC curve close to the 45° line, along which the TPR equals the FPR, and an AUC of 0.5; values above 0.5 mean that the series runs higher on crisis dates, and values below 0.5 that it runs lower. Because the AUC depends only on ranks, it is invariant to the base-100 convention and to monotone transformations such as squaring, and it does not depend on how many crisis dates the calendar contains. It fits no model and imposes no lag, and the labels come from Appendix B, Table 7, dated from event chronologies rather than from market data. Appendix A.4 sets out these definitions and the inference. Appendix B, Table 9 reports the results window by window and pooled by type of episode.

Pooled over all 234 crisis dates, EPU_t scores 0.540, barely above the uninformative value. That figure mixes two kinds of episodes. Splitting the windows by their type, recorded in Appendix B, Table 7 without using any series, resolves it: EPU_t scores 0.938 on the 84 dates of the two economic windows and 0.318 on the 150 dates of the six political ones. Panel A shows the pattern window by window: 0.960 for the COVID-19 emergency C3 and 0.806 for the fuel-price episode C6, against 0.098 for the Merino interregnum C4 and 0.263 for the post-Castillo protests C8. EPU_t is close to a perfect detector of the economic windows and an inverted detector of the political windows as a group; only C1 (0.600) lies above the uninformative value. By the symmetry of Eq. (A.11), its political-window score corresponds to 0.682 for the reversed series, which is weaker than its economic-window score.

Is the exercise circular? The index is built from discourse that reacts to the same events that date the windows, so uniformly high scores would prove little. The scores are not uniformly high: EPU_t falls to 0.098 in C4, far below the uninformative value. It separates episodes by their economic content; it does not echo the calendar it is scored against. Because the trailing window delays the index's response, moving the start of each window earlier adds dates on which the index has not yet reacted. Extending each crisis start backward by five days lowers the economic-window AUC of EPU_t from 0.938 to 0.901 and the political-window AUC of PU_t excluding C5 from 0.891 to 0.836; discrimination remains substantial under this change.

Appendix C, Figure 12 shows the same results as curves. Panel (a) pools the economic windows and panel (b) the political ones, each against the same 954 calm dates. The solid black line is the 45° line, along which a crisis date is as likely as a calm date to exceed any threshold. A curve above

it means that, for every share of calm dates wrongly flagged, the series flags a larger share of crisis dates, and the area between the curve and the line measures that advantage. A curve below it ranks crisis dates below calm dates and carries as much information as its mirror image. In panel (a), EPU_t and the VIX rise steeply near the origin: at low false positive rates they already flag most economic-crisis dates, so both identify the pandemic, which supplies 72 of the 84 economic dates, with few false alarms; for the fuel-price episode C6 the VIX is much weaker (0.566 in Panel A of Appendix B, Table 9). PU_t and r_t^2 run closer to the line. In panel (b) the ordering changes: PU_t is the only series clearly above the line, the VIX and r_t^2 stay around it, and EPU_t runs below it. The curves cross, so no single threshold summarizes the comparison, which is why we report the area under each curve.

This is a property of the construction. EPU_t counts a tweet only in the $E \cap P \cap U$ cell of Section 3: a constitutional crisis with no immediate economic content does not qualify, and the flood of political tweets raises the fortnightly denominator without raising the strict count, so the rate falls. PU_t, which requires only $P \cap U$, is the mirror image, and its pooled score of 0.686 on the political windows understates it for the same reason: the contested general election C5 spans 76 of the 150 political trading dates and returns 0.487, so half that pooled sample is a single window on which PU_t is uninformative. Dropping C5 leaves 74 political dates on which PU_t scores 0.891 against 0.361 for EPU_t, a difference of 0.530.

Whether the two indices differ requires a test that accounts for the fact that both are scored on the same dates, so that their AUC estimates are correlated. The test of DeLong et al. (1988) does so through the covariance of the two indices' placement values (Appendix A.4), and Panel C of Appendix B, Table 9 reports its z statistic. It treats dates as independent, however, which the persistence in Appendix B, Table 5 rules out, so we base inference on a circular block bootstrap (Politis and Romano, 1992) and report the DeLong z only as a conventional reference. For the political windows excluding C5, the difference has a DeLong z of 17.35 and a bootstrap 95% interval of [0.352, 0.681]; on the economic windows the difference is -0.241, interval $[-0.361, -0.104]$. The sign reverses with the character of the episode, and the two scores are near mirrors: 0.938 for EPU_t on economic dates and 0.891 for PU_t on political dates. The indices are not competing measures of one quantity; they partition the crisis calendar between them.

The benchmarks place this reading. The VIX scores 0.893 on the economic windows and 0.481 on the political ones, rising only to 0.540 once C5 is dropped, against 0.891 for PU_t: it registers the pandemic, a global shock, but is close to blind to Peru's domestic political episodes. The squared return r_t^2 scores 0.585 on all windows and 0.532 on the political windows without C5, discriminating weakly and indiscriminately. The conditional variance $\hat{\sigma}^2_{\text{BVL},t}$ separates the economic windows (0.871) and scores 0.684 on the political ones, close to PU_t, but this reflects C4 (0.950) and C5 (0.862), when Lima's stock market was turbulent; without C5 it falls to 0.501. No market series separates the remaining political windows from calm days, and the text indices fill that gap. C5, 76 trading dates that combine an acute fortnight with three quieter months, is close to the uninformative value for PU_t (0.487) and the VIX (0.423), and EPU_t ranks it below calm dates (0.275); this reflects the coarse dating of Appendix B, Table 7 rather than a property of the indices.

5.4 Conditional Variance

The Gaussian GARCH(1, 1) fitted to the 1 188 BVL returns yields $\hat{\alpha} = 0.115$ and $\hat{\beta} = 0.884$, with persistence $\hat{\alpha} + \hat{\beta} = 0.9994$ (Appendix B, Table 6). The implied half-life of a variance shock, about 1 170 trading days, is too long to be estimated reliably, so we do not interpret it as a forecast horizon; under Student-t innovations persistence is 0.988 and the half-life 59 days. The Ljung–Box test at ten lags does not reject the absence of remaining dependence in the squared standardized

residuals ($p = 0.70$ under Gaussian innovations and $p = 0.19$ under Student-t innovations), and the two fitted variance series correlate at 0.994.

The trading-day panel matters for these estimates. Appendix C, Figure 13 compares $\hat{\sigma}^2_{\mathrm{BVL},t}$ with the conditional variance from the same model estimated on the calendar panel, in which non-trading dates carry linearly interpolated prices. Interpolation spreads each price move over several days and lowers the measured variance, and it changes the fitted parameters (Appendix B, Table 6, sensitivity row).

Appendix C, Figure 14 places EPU_t, VIX_t, and $\hat{\sigma}^2_{\mathrm{BVL},t}$ on a common standardized scale. The three rise together most clearly in C3; elsewhere their responses differ, and $\hat{\sigma}^2_{\mathrm{BVL},t}$ also rises in C4 and during C5, when EPU_t stays low. Appendix B, Table 10 reports the correlations. EPU_t correlates with $\hat{\sigma}^2_{\mathrm{BVL},t}$ at 0.285 over the full panel (HAC standard error 0.093), 0.292 on crisis dates, and 0.235 on calm dates, while the correlation between VIX_t and $\hat{\sigma}^2_{\mathrm{BVL},t}$ is 0.636. The text index is therefore more closely related to the VIX than to the variance of Lima returns, and the two market measures are more closely related to each other than either is to the index. These moderate associations are consistent with complementary measures of uncertainty.

5.5 Tweet Volume and the Precision of the Index

EPU_t is an estimated share, so its precision depends on how many tweets enter each window. Let $\mathcal{N}_t = \sum_{s=t-13}^{t} N_s$ be the number of classified tweets posted in the fourteen calendar days that end on day t, including t; it is the denominator of the baseline rate ρ_t in Eq. (3). Under the binomial approximation of Eq. (5), more tweets mean a more precise share: the standard error falls with $\mathcal{N}_t^{-1/2}$, so quadrupling the volume halves it. Expressed in index points, $100\,\mathrm{se}_t/\bar{\rho}$, the standard error has a median of 11.8. It is about 20 points in 2018 and 2019, when the median window holds about 210 tweets, and below 6 points in 2022, when it holds about 3 900 (Appendix C, Figure 15, panel a).

The observed short-run variability of the index follows the same gradient. Across deciles of $\mathcal{N}_t$, the mean absolute daily change $|\Delta\mathrm{EPU}_t|$ falls from 10.9 index points in the lowest-volume decile to 3.4 in the highest (panel b), and across the 1 834 daily observations its correlation with the binomial standard error is 0.44. These associations are compatible with a measurement-error component that shrinks with volume. They do not identify how much of the observed variation is measurement error: volume is not assigned at random, since it trends upward over the sample and rises during crises, when genuine uncertainty also changes, and the binomial approximation ignores the dependence among tweets by the same account. We therefore compare the level of EPU_t, a fourteen-day average whose sampling error is small relative to the episodes of Section 5.2, with the market benchmarks, and we use the variability of the index only as a diagnostic of its precision.

5.6 An Example: Central Bank Intervention in the Foreign Exchange Market

As an example of how the indices can inform policy analysis, we ask whether political uncertainty is associated with the volume of foreign-exchange intervention by the BCRP. The central bank intervenes, mostly through spot operations and foreign-exchange swaps, to smooth fluctuations in the value of the Sol, and episodes of political uncertainty can raise the demand for dollars and with it the pressure on the exchange rate. The exercise illustrates a use of the indices; it is not a model of the BCRP's reaction function. Appendix B, Table 11 and Appendix C, Figure 16 report the

VAR of Section 4.5.[3] For PU_t, the BIC selects $p = 3$ on the full sample and $p = 2$ in the crisis blocks.

A one-standard-deviation PU_t shock (about 5.6 index points) is associated with USD 10.4 million more gross intervention on the same day, with a 95% interval of $[+0.4, +20.3]$. The response reverses the next day (-12.9 million), so the point estimates imply no increase in intervention cumulated over the two days (-2.6 million). Past PU_t nevertheless predicts intervention (Granger $p = 0.016$). Within the crisis blocks, the same-day response is USD 13.2 million, interval $[+1.6, +27.2]$; its larger point estimate does not establish a difference between samples. Absolute net spot purchases increase by USD 2.4 million on impact, interval $[+0.5, +4.4]$, but their Granger test does not reject ($p = 0.874$).

Replacing PU_t with EPU_t on the same dates gives $p = 2$, a same-day response of USD 6.0 million, interval $[-3.9, +16.0]$, and Granger $p = 0.047$. Within crisis blocks, $p = 1$ and the response is USD 6.9 million, interval $[-6.4, +25.2]$. On the full sample the two intervals have almost the same width; PU_t has the larger same-day response and the only interval that excludes zero, while both indices have full-sample predictive content for the volume of intervention. Because $|INT_t|$ measures volume, these results do not say whether the BCRP sold or bought dollars, and the same-day response rests on the recursive ordering; they establish an association, not structural causality.

5.7 What the Index Contributes

The two indices complement market measures by distinguishing economic uncertainty from political rupture in domestic discourse. EPU_t is most elevated during economic crises; PU_t captures political episodes such as C1, C7, and C8 and has predictive content for the volume of intervention. The VIX measures the option-implied volatility of United States equities, while r_t^2 and $\hat{\sigma}^2_{\mathrm{BVL},t}$ measure the variance of Peruvian equity returns. Moderate correlations are consistent with these different sources and coverage.

6 Conclusions

We construct daily EPU_t and PU_t indices from 213 249 messages posted by 97 influential X accounts. Contextual classification and volume-weighted aggregation distinguish economic and political uncertainty from political uncertainty more broadly. Appendix C, Figure 17 summarizes the findings.

EPU_t discriminates economic crises well (AUC 0.938), while PU_t identifies political ruptures more clearly (AUC 0.891 excluding the contested-election window C5). EPU_t correlates with the VIX at 0.434 overall and 0.648 in crises, and with the conditional variance of BVL returns at 0.285. The correlation between daily changes in EPU_t and the VIX is near zero (-0.015), indicating that the association mainly reflects variation across episodes. The precision of the index rises with tweet volume, so readings from the thin early part of the sample are noisier.

In the example of Section 5.6, PU_t predicts the volume of BCRP foreign-exchange intervention (Granger $p = 0.016$). Under the stated Cholesky ordering, its same-day response is USD 10.4 million, with a 95% interval of $[+0.4, +20.3]$, and it is reversed the next day; the crisis estimate is USD 13.2 million, interval $[+1.6, +27.2]$. These results establish an association, not structural causality. The indices provide a domestic complement to financial measures, subject to the selected accounts, classification error, and the limited number of crisis episodes.

[3] BCRPData series PD04658MD and PD04659MD, matched on 1 183 trading dates. Gross intervention $|INT_t|$ averages USD 37.9 million per day; absolute net spot purchases $|COM_t|$ average USD 10.3 million.

Appendix

A Econometric Background

This appendix provides the unit-root test, the HAC inference, the GARCH properties, and the ROC statistics used in Sections 4 and 5. Appendices B and C contain the tables and figures.

A.1 Unit-Root Test

Appendix B, Table 5 reports the augmented Dickey–Fuller test (Dickey and Fuller, 1979; Said and Dickey, 1984), computed with the `adfuller` routine of the Python library `statsmodels`. For a series x_t, the test estimates the regression:

$$\Delta x_t = c + \alpha\, x_{t-1} + \sum_{k=1}^{q} \delta_k\, \Delta x_{t-k} + u_t, \tag{A.1}$$

where c is a constant and the lagged differences absorb serial correlation in u_t. The number of lags q minimizes the BIC among $q = 0, \ldots, 23$. The null of a unit root, $H_0 : \alpha = 0$, is tested against the stationary alternative $\alpha < 0$ with the t statistic of $\hat{\alpha}$, whose Dickey–Fuller critical values for this sample are -3.44, -2.86, and -2.57 at the 1%, 5%, and 10% levels.

A.2 HAC Inference for a Correlation

The Pearson correlation equals the slope between standardized variables, $\tilde{e}_t = (\mathrm{EPU}_t - \overline{\mathrm{EPU}})/s_{\mathrm{EPU}}$ and $\tilde{b}_t = (b_t - \bar{b})/s_b$, where s_{EPU} and s_b are sample standard deviations computed with divisor T. Its sampling variance must also account for estimating those means and standard deviations. Applying the delta method to the correlation gives the influence score:

$$g_t(\rho) = \tilde{e}_t \tilde{b}_t - \frac{\rho}{2}\left(\tilde{e}_t^2 + \tilde{b}_t^2\right). \tag{A.2}$$

Evaluating it at $\hat{\rho}$ gives $\hat{g}_t$. Under weak dependence and finite moments, the asymptotic variance of $\hat{\rho}$ is S/T, where S is the long-run variance of g_t, the sum of its autocovariances at all lags.

The Newey and West (1987) estimator of S down-weights the sample autocovariances $\hat{\gamma}_j$ of $\hat{g}_t$ with the weights of Bartlett (1950):

$$\hat{S}(L) = \hat{\gamma}_0 + 2\sum_{j=1}^{L}\left(1 - \frac{j}{L+1}\right)\hat{\gamma}_j, \qquad \hat{\gamma}_j = \frac{1}{T}\sum_{t=j+1}^{T} \hat{g}_t \hat{g}_{t-j}, \tag{A.3}$$

where L is the bandwidth. The estimated variance of the correlation is therefore:

$$\widehat{\mathrm{Var}}(\hat{\rho}) = \frac{\hat{S}(L)}{T}. \tag{A.4}$$

Bartlett weighting makes this estimate nonnegative. The classical standard error $(1 - \hat{\rho}^2)/\sqrt{T}$ is reported only as an independent Gaussian reference.

The adaptive bandwidth reported as L^A in Appendix B, Table 8 is computed from the lag-one sample autocorrelation $\hat{a}$ of $\hat{g}_t$:

$$L^A = \max\left\{1, \min\left[T-1, \left\lceil 4(T/100)^{2/9}\left(\frac{|\hat{a}|}{1-\hat{a}^2}\right)^{2/9}\right\rceil\right]\right\}.$$

More persistent scores, with $|\hat{a}|$ closer to one, receive longer bandwidths. Fixed bandwidths $L \in \{7, 30, 60, 100\}$ provide a sensitivity check.

A.3 The GARCH(1,1) Model of BVL Returns

This subsection collects the properties of the GARCH$(1,1)$ of Eqs. (11) and (12) that underlie the statistics of Appendix B, Table 6. Throughout, $\mathcal{F}_{t-1}$ is the information available at the close of the previous trading date and $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$ is the conditional variance of Section 4.4; its square root σ_t is the conditional volatility. The recursion of Eq. (12) is initialized at the sample variance of the innovations, and by induction σ_t^2 depends only on returns observed before t, as a conditional variance must.

Unconditional variance. Taking unconditional expectations in Eq. (12) requires $\mathbb{E}[\varepsilon_{t-1}^2]$. Because σ_{t-1}^2 is known at $t-2$ and z_{t-1} is independent of the past with $\mathbb{E}[z_{t-1}^2] = 1$, the law of iterated expectations gives:

$$\mathbb{E}\Big[\varepsilon_{t-1}^2\Big] \;=\; \mathbb{E}\Big[\mathbb{E}\Big[\sigma_{t-1}^2\, z_{t-1}^2 \mid \mathcal{F}_{t-2}\Big]\Big] \;=\; \mathbb{E}\Big[\sigma_{t-1}^2\, \mathbb{E}\Big[z_{t-1}^2 \mid \mathcal{F}_{t-2}\Big]\Big] \;=\; \mathbb{E}\Big[\sigma_{t-1}^2\Big]\,.$$

If the process is covariance-stationary, $\mathbb{E}[\sigma_t^2] = \mathbb{E}[\sigma_{t-1}^2] = \bar{\sigma}^2$, so $\bar{\sigma}^2 = \omega + (\alpha+\beta)\,\bar{\sigma}^2$, and the unconditional variance is:

$$\bar{\sigma}^2 \;=\; \frac{\omega}{1-\alpha-\beta}. \tag{A.5}$$

This expression requires $\alpha+\beta < 1$. As persistence approaches one, the unconditional variance grows without bound, which is why estimates close to that boundary, such as the Gaussian estimate in Appendix B, Table 6, are hard to interpret; the estimation imposes $\alpha+\beta \le 0.9999$.

Half-life. Iterating the conditional expectation of Eq. (12) forward gives the h-step variance forecast:

$$\mathbb{E}\Big[\sigma_{t+h}^2 \mid \mathcal{F}_t\Big] - \bar{\sigma}^2 \;=\; (\alpha+\beta)^{\,h-1}\left(\sigma_{t+1}^2 - \bar{\sigma}^2\right), \qquad h \ge 1.$$

Deviations of expected variance from its long-run level therefore decay geometrically at rate $\alpha+\beta$, and a variance shock has a half-life of $\ln(1/2)/\ln(\alpha+\beta)$ trading days, the "variance half-life" of Appendix B, Table 6.

Filter memory. Repeated substitution for $\beta\sigma_{t-1}^2$ yields an ARCH representation of infinite order:

$$\sigma_t^2 \;=\; \frac{\omega}{1-\beta} + \alpha \sum_{j=0}^{\infty} \beta^{\,j}\, \varepsilon_{t-1-j}^2.$$

The squared innovation $j+1$ days back receives weight $\alpha\beta^j$, so the weights decay geometrically and $1/(1-\beta) = \sum_{j\ge 0}\beta^j$ measures the effective length of the filter's memory; Appendix B, Table 6 reports it as "filter memory". The single pair (α,β) thus encodes a complete memory profile, which is what makes the $(1,1)$ order sufficient in practice.

Estimation. Under Gaussian innovations, $\varepsilon_t \mid \mathcal{F}_{t-1} \sim \mathrm{N}(0,\sigma_t^2)$, so the joint density of the returns factorizes into one-step conditional densities, and the log-likelihood of $\theta = (\mu,\omega,\alpha,\beta)$ is:

$$\ell(\theta) \;=\; -\frac{1}{2}\sum_{t=1}^{T}\left[\ln(2\pi) + \ln\sigma_t^2(\theta) + \frac{\varepsilon_t(\theta)^2}{\sigma_t^2(\theta)}\right]. \tag{A.6}$$

For each trial parameter vector θ, the innovations are $\varepsilon_t(\theta) = r_t - \mu$, their sample variance initializes the recursion of Eq. (12), and the resulting $\sigma_t^2(\theta)$ enter Eq. (A.6), which is maximized numerically. The Student-t version replaces the Gaussian density by a standardized Student-t density with $\nu > 2$ degrees of freedom; it changes the likelihood, not the variance recursion.

A.4 The Area Under the ROC Curve

This subsection defines the statistics of Appendix B, Table 9 and explains their sampling uncertainty. Let $\mathcal{K}$ contain the n_k crisis dates and $\mathcal{C}$ the n_c calm dates of a comparison, with crisis labels taken from Appendix B, Table 7. For a threshold c, classify date t as a crisis when the series X_t exceeds c. At the population level, the true and false positive rates are:

$$\mathrm{TPR}(c) = \Pr\left(X_t > c \mid t \in \mathcal{K}\right), \qquad \mathrm{FPR}(c) = \Pr\left(X_t > c \mid t \in \mathcal{C}\right). \tag{A.7}$$

The ROC curve plots $\mathrm{TPR}(c)$ vertically against $\mathrm{FPR}(c)$ horizontally as the threshold varies; lowering c moves the curve from $(0,0)$ to $(1,1)$. Write $\mathrm{ROC}(u)$ for its height at false positive rate u. For the empirical curve, adjacent threshold points are joined linearly, including when observations are tied. The area under the curve is:

$$\mathrm{AUC} = \int_0^1 \mathrm{ROC}(u)\,\mathrm{d}u. \tag{A.8}$$

The 45° line $\mathrm{ROC}(u) = u$ has area $1/2$ and corresponds to no discrimination; a series that is higher on every crisis date than on every calm date has area 1.

Let X^K be the value of the series on a randomly drawn crisis date and X^C its value on a randomly drawn calm date, the two draws independent. The area then has a probabilistic interpretation:

$$\mathrm{AUC} = \Pr\left(X^K > X^C\right) + \tfrac{1}{2}\Pr\left(X^K = X^C\right), \tag{A.9}$$

the probability that the series ranks a crisis date above a calm date, with ties split. It is estimated by the two-sample U-statistic of Mann and Whitney (1947):

$$\widehat{\mathrm{AUC}} = \frac{1}{n_k n_c}\sum_{t\in\mathcal{K}}\sum_{s\in\mathcal{C}}\psi\left(X_t, X_s\right), \qquad \psi(a,b) = \mathbf{1}\{a > b\} + \tfrac{1}{2}\mathbf{1}\{a = b\}, \tag{A.10}$$

which is the Mann–Whitney statistic divided by $n_k n_c$.

Three properties matter for the interpretation of Appendix B, Table 9. First, $\widehat{\mathrm{AUC}}$ depends on the data only through the ranks of the pooled sample, so it is invariant to any strictly increasing transformation of X_t. The base-100 rescaling of Eq. (6) therefore cannot affect it, and neither can squaring a positive series; nothing in the table turns on units. Second, the estimand conditions on the two populations separately, so it does not depend on how many crisis dates the calendar contains relative to calm ones. A short window is not thereby penalized, which matters here because the eight windows range from eight to seventy-six trading dates. Third, reversing the sign of the series reverses the area:

$$\mathrm{AUC}\left(-X\right) = 1 - \mathrm{AUC}\left(X\right), \tag{A.11}$$

so $\frac{1}{2}$ is the uninformative value and the discriminating content of a series is $\left|\mathrm{AUC} - \frac{1}{2}\right|$. A value below one half is not a failed detector but an inverted one, carrying exactly as much information as its mirror image. For EPU_t, the political-window AUC of 0.318 therefore corresponds to 0.682 after reversing its sign. This inverse ranking is weaker than the AUC of 0.938 on the economic windows; the direction and the strength of discrimination must be distinguished.

To compare two series scored on the same dates, Panel C of Appendix B, Table 9 uses the test of DeLong et al. (1988). Because Eq. (A.10) is a two-sample U-statistic of degree $(1,1)$, its variance follows from the Hájek projection. Define the placement values:

$$V_t = \frac{1}{n_c}\sum_{s\in\mathcal{C}}\psi\left(X_t, X_s\right),\ t \in \mathcal{K}, \qquad W_s = \frac{1}{n_k}\sum_{t\in\mathcal{K}}\psi\left(X_t, X_s\right),\ s \in \mathcal{C}, \tag{A.12}$$

the fraction of calm dates that a given crisis date outranks, and the fraction of crisis dates that outrank a given calm date. Both average to $\widehat{\mathrm{AUC}}$. Let S_{10} and S_{01} be the sample variances of V_t and W_s, computed with denominators $n_k - 1$ and $n_c - 1$, respectively. The estimator of DeLong et al. (1988) for the variance of one area is:

$$\widehat{\mathrm{Var}}\left(\widehat{\mathrm{AUC}}\right) = \frac{S_{10}}{n_k} + \frac{S_{01}}{n_c}. \tag{A.13}$$

For two series scored on the same dates, the placement values are formed for both, and S_{10} and S_{01} become 2×2 covariance matrices across series. The variance of the difference between the two areas is then $\ell' \left(S_{10}/n_k + S_{01}/n_c\right) \ell$ with $\ell = (1, -1)'$, and the DeLong z is the difference divided by the square root of this variance.

The DeLong covariance accounts for the fact that both indices are observed on the same dates. It does not account for serial dependence across dates, which matters here because successive fourteen-day index windows overlap. We therefore use the circular block bootstrap of Politis and Romano (1992), described in the note to Appendix B, Table 9, resampling the indices and labels jointly in blocks after pooling dates by crisis status, in chronological order within each group. This retains local within-group dependence, but not all original calendar adjacencies. Its standard errors are 2.8 to 3.6 times the DeLong estimates in this sample. The DeLong z statistic is reported as an independence-based reference; inference uses the bootstrap intervals.

B Tables

Table 1. Complete List of X Accounts (97)

Username	Name	Category
RosanaCuevaM	Rosana Cueva	Journalist
NicolasLucar	Nicolás Lucar	Journalist
PolloFarsantePe	Beto Ortiz	Journalist
MilagrosLeivaG	Milagros Leiva	Journalist
DeltaMdelta	Mónica Delta	Journalist
MavilaHuertasC	Mavila Huertas	Journalist
SolCn	Sol Carreño	Journalist
VertizPamela	Pamela Vértiz	Journalist
claudia_cisneros	Claudia Cisneros	Journalist
recisneros	Renato Cisneros	Journalist
larryportera	Paola Ugaz	Journalist
tuesta	Fernando Tuesta Soldevilla	Journalist
Federicoagust	Federico Salazar B.	Journalist
julianaoxenford	Juliana Oxenford	Journalist
mariateguiperu	Aldo Mariátegui	Journalist
JaimeChincha	Jaime Chincha	Journalist
VeroLinaresC	Verónica Linares	Journalist
tenoriopedro	Pedro Tenorio	Journalist
JBCPERU	José Barba Caballero	Journalist
jdealthaus	Jaime de Althaus	Journalist
ensustrece	Hildebrandt en sus trece	Journalist
BaylyOficial	Jaime Bayly	Journalist
jctafur	Juan Carlos Tafur	Journalist
cparedesr	Carlos Paredes	Journalist
PCaterianoB	Pedro Cateriano B.	Politician
Carlos_Bruce	Carlos Bruce	Politician
JuanSheput	Juan Sheput	Politician
VLADIMIR_CERRON	Vladimír Cerrón	Politician
anibaltorresv	Aníbal Torres V.	Politician
pedrofrancke	Pedro Francke	Politician
Ollanta_HumalaT	Ollanta Humala Tasso	Politician
sigridbazan	Sigrid Bazán	Politician
MirtyVas	Mirtha Vásquez	Politician
FlorPabloMedina	Flor Pablo Medina	Politician
rlopezaliaga1	Rafael López Aliaga	Politician
patarevalo	Patricia Arévalo	Politician
George_Forsyth	George Forsyth	Politician
JorgeMunozPe	Jorge Muñoz	Politician
DanielUrresti1	Daniel Urresti	Politician
VictorAndresGB	Víctor Andrés García Belaunde	Politician
KeikoFujimori	Keiko Fujimori	Politician
MartinVizcarraC	Martín Vizcarra	Politician

Username	Name	Category
JorgeDCG	Jorge del Castillo	Politician
Mauriciomulder	Mauricio Mulder	Politician
nidiavilchez	Nidia Vílchez	Politician
FSagasti	Francisco Sagasti	Politician
AlbertoBelaunde	Alberto de Belaunde	Politician
MaricarmenAlvaP	Maricarmen Alva	Politician
PattyChirinos1	Patricia Chirinos	Politician
adrianatudelag	Adriana Tudela Gutiérrez	Politician
CesarAcunaP	César Acuña Peralta	Politician
MesiasGuevara	Mesías Guevara	Politician
vozdelatierra	Marco Arana	Politician
MarisaGlave	Marisa Glave	Politician
GuilleBermejoR	Guillermo Bermejo Rojas	Politician
robertochiabra	Roberto Chiabra	Politician
RichardArcePeru	Richard Arce	Politician
Alm_Montoya	Jorge Montoya	Politician
AlejandroCavero	Alejandro Cavero	Politician
HectorValer_PER	Héctor Valer Pinto	Politician
RosselliAmuruz	Rosselli Amuruz	Politician
HDeSotoPeru	Hernando de Soto	Politician
UrsulaLetonaP	Ursula Letona	Politician
AlvaroVargasLl	Álvaro Vargas Llosa	Politician
AlbertoOtarolaP	Alberto Otárola	Politician
HPerezDeCuellar	Hania Pérez de Cuéllar	Politician
alosegura	Alonso Segura Vasi	Economist
BustamantePao18	Paola Bustamante S.	Economist
aethorne	Alfredo E. Thorne	Economist
WaldoMendozaB	Waldo Mendoza Bellido	Economist
JuanManuelGarc3	Juan Manuel García C.	Economist
CarlosAnderso1	Carlos A. Anderson	Economist
hugonopo	Hugo Ñopo	Economist
tuestadavid	David Tuesta	Economist
pablo_lavado	Pablo Lavado	Economist
OswaldoMolinaC	Oswaldo Molina	Economist
CParodiT	Carlos Parodi Trece	Economist
psecadae	Pablo Secada	Economist
DWinkelried	Diego Winkelried	Economist
romerocaroperu	Manuel Romero Caro	Economist
manupulgarvidal	Manuel Pulgar Vidal	Economist
BancalariA	Antonella Bancalari	Economist
eljorobado	Carlos Meléndez	Political analyst
DargentEduardo	Eduardo Dargent	Political analyst
DelaPuenteJuan	Juan De la Puente	Political analyst
lauermirko	Mirko Lauer	Political analyst
rmapalacios	Rosa María Palacios	Political analyst
Frospigliosi	Fernando Rospigliosi	Political analyst

Username	Name	Category
BetoAdrianzen	Alberto Adrianzén	Political analyst
ocram	Marco Sifuentes	Political analyst
brunoschaaf	Bruno Schaaf	Political analyst
luisbenaventegi	Luis Benavente	Political analyst
camcesar	César Campos	Political analyst
AlfredoMTorres	Alfredo M. Torres G.	Political analyst
gonza_banda	Gonzalo Banda	Political analyst
Roque_Benavides	Roque Benavides	Private sector
CayetanaAljovin	Cayetana Aljovín	Private sector

Table 2. Label Combinations (E, P, U)

E	P	U	Interpretation	Count
0	0	0	Unrelated to economics, politics, or uncertainty	81 353
1	0	0	Economic content without politics or uncertainty	2 051
0	1	0	Political content without economics or uncertainty	19 280
0	0	1	Uncertainty not linked to economics or politics	23 144
1	1	0	Economic and political content, no uncertainty	9 674
1	0	1	Economic uncertainty	1 318
0	1	1	Political uncertainty	59 978
1	1	1	Economic and political uncertainty (strict cell)	16 451

Note: the eight cells partition the 213 249 classified tweets and sum to that total.

Table 3. Descriptive Statistics

Statistic	Value
Total tweets	213 249
Unique accounts	97
Date range	2018-01-23 to 2023-01-31
Calendar days	1 835
Daily N_t: Mean / Median / Max	116.21 / 56 / 1 016
Total strict tweets ($\sum_t Y_t$)	16 451
Days with $Y_t > 0$	1 462
Y_t mean (cond. on $Y_t > 0$)	11.25
Y_t maximum	82

Note: N_t is the daily tweet volume and Y_t the daily count of strict tweets ($E \cap P \cap U$); the conditional mean is taken over the 1 462 days with $Y_t > 0$.

Table 4. Comparison of the Index and the Three Benchmarks

	EPU_t	VIX_t	r_t^2	$\hat{\sigma}^2_{\text{BVL},t}$
Underlying data	Classified tweets	S&P 500 index options	S&P/BVL index prices	S&P/BVL index returns
Geographic origin	Peru	United States	Peru	Peru
Quantity	Share of uncertain discourse	Implied volatility (standard deviation)	Squared daily log return (variance proxy)	Conditional variance (GARCH)
Units	Index, sample mean = 100	Annualized percentage points	Squared decimal return	Squared decimal return
Horizon	Trailing 14 calendar days	Next 30 calendar days	Current trading day, realized	Current trading day, forecast from past returns
Price-anchored	No	Yes	Yes	Yes
Availability	Calendar days	Trading days	Trading days	Trading days

Note: the VIX is option-implied from Standard & Poor's 500 (S&P 500) options and measures volatility, a standard deviation; r_t^2 and $\hat{\sigma}^2_{\text{BVL},t}$ measure the variance of the BVL log return r_t. $\hat{\sigma}^2_{\text{BVL},t}$ is the conditional variance from the GARCH$(1,1)$ of Section 4.4. The index reflects uncertainty as it is expressed in discourse. Price-anchored indicates whether the measure derives from a traded price.

Table 5. Persistence Diagnostics

Series	ADF t	Lags	$\hat{\rho}_1$	Half-life (days)
EPU_t	-3.57^a	10	0.969	21.8
VIX_t	-3.91^a	1	0.963	18.4
r_t^2	-12.11^a	3	0.182	0.4
$\hat{\sigma}^2_{\mathrm{BVL},t}$	-4.93^a	0	0.960	16.8

Note: $T = 1\,188$ trading dates with a valid daily return (Section 4.1). Here $r_t = \log(P_t/P_{t-1})$, so r_t^2 is the squared log return, not the logarithm of a squared return. ADF is the t statistic of the augmented Dickey–Fuller regression of Eq. (A.1), with a constant and the number of lagged differences (Lags) chosen by the BIC among 0 to 23. Superscripts a, b, and c indicate rejection of the unit-root null at 1%, 5%, and 10%, respectively (critical values -3.44, -2.86, -2.57). $\hat{\rho}_1$ is the first-order autocorrelation and the half-life is $\ln(1/2)/\ln\hat{\rho}_1$, in trading days. All four series reject the unit root at 1%. The high autocorrelations of EPU_t, VIX_t, and $\hat{\sigma}^2_{\mathrm{BVL},t}$ nevertheless indicate strong persistence. For r_t^2, dependence decays within a day.

Table 6. GARCH(1, 1) Estimates for the BVL Return r_t

Innovation distribution	Gaussian	Student-t
Mean equation	Constant	Constant
$\hat{\omega}$	0.0164	0.0238
$\hat{\alpha}$	0.1154	0.0818
$\hat{\beta}$	0.8840	0.9066
$\hat{\nu}$	—	4.24
Persistence $\hat{\alpha} + \hat{\beta}$	0.9994	0.9884
Variance half-life (days)	1 169.5	59.3
Filter memory $1/(1 - \hat{\beta})$	8.6	10.7
Log-likelihood	−1 768.7	−1 709.4
Observations	1 188	1 188
$\hat{\rho}_1(\hat{z}_t^2)$	0.021	0.055
Ljung–Box(10) p-value	0.70	0.19
Sensitivity check: Gaussian model on the calendar panel with interpolated prices		
$\hat{\alpha}$ / $\hat{\beta}$ / persistence	0.0799 / 0.9180 / 0.9979	
Variance half-life / T / Ljung–Box p	332.2 days / 1 834 / 0.06	

Note: GARCH(1, 1) of Eqs. (11)–(12), estimated by maximum likelihood on the trading-day panel. It is the only GARCH model in the paper and is applied to r_t, never to r_t^2 or to the text indices. The model is estimated on $100\, r_t$, so $\hat{\omega}$ is in squared percentage points; the filtered variance $\hat{\sigma}^2_{\mathrm{BVL},t}$ is converted to squared decimal units. The Student-t column re-estimates the same model with a standardized t innovation; $\hat{\nu}$ close to 4 means that the fourth moment is barely finite, so the Gaussian fit is a quasi-likelihood. Neither estimate reaches the imposed persistence bound $\hat{\alpha} + \hat{\beta} \leq 0.9999$. $\hat{\rho}_1(\hat{z}_t^2)$ is the first-order autocorrelation of the squared standardized residuals. The sensitivity rows use calendar dates, including linearly interpolated non-trading prices, in place of the trading-day panel.

Table 7. Crisis Windows in Peru, 2018–2023

ID	Start	End	Days	Type	Episode
C1	2018-03-12	2018-04-06	26	Political	Fall of Kuczynski; transition to Vizcarra
C2	2019-09-27	2019-10-15	19	Political	Dissolution of Congress
C3	2020-03-15	2020-06-30	108	Economic	COVID-19 emergency and lockdown
C4	2020-11-09	2020-11-18	10	Political	Vizcarra vacancy; Merino interregnum
C5	2021-04-08	2021-07-28	112	Political	Contested general election
C6	2022-03-28	2022-04-12	16	Economic	Fuel-price protests; Lima curfew
C7	2022-12-07	2022-12-20	14	Political	Castillo self-coup and removal
C8	2022-12-20	2023-01-31	43	Political	Post-Castillo protests
Union of windows			347	18.9% of the 1 835 sample days	

Note: windows are dated from the historical record, not market volatility, keeping the crisis/calm split exogenous to the benchmarks. Inclusive calendar days. Type records the origin of the episode and uses no information from any series. C7 and C8 share 2022-12-20, counted once in the union ($26 + 19 + 108 + 10 + 112 + 16 + 14 + 43 = 348$ window days; union = 347). C8 is right-censored by the sample end. Edges flagged as judgment calls are checked by shifting all windows by ± 5 days.

Table 8. Static Co-Movement of EPU_t with VIX_t and r_t^2

				Standard error of $\hat{\rho}$					
Sample	T	$\hat{\rho}$	$\hat{\rho}_S$	Classical	$L=7$	$L=30$	$L=60$	$L=100$	L^A
Panel A. Benchmark $b_t = \mathrm{VIX}_t$									
Full	1 188	0.434	0.408	0.024	0.077	0.095	0.093	0.087	0.085
Crisis	234	0.648	0.658	0.038	0.098	0.079	0.065	0.057	0.096
Calm	954	0.261	0.331	0.030	0.072	0.093	0.097	0.096	0.074
Panel B. Benchmark $b_t = r_t^2$									
Full	1 188	0.050	0.124	0.029	0.038	0.033	0.032	0.034	0.037
Crisis	234	0.010	0.029	0.065	0.070	0.077	0.084	0.074	0.066
Calm	954	0.047	0.154	0.032	0.044	0.050	0.055	0.057	0.042

Note: Pearson $\hat{\rho}$ of Eq. (8) and Spearman $\hat{\rho}_S$ of Eq. (9) between EPU_t and each benchmark, on the 1 188 trading dates of Section 4.1. Classical is $\left(1-\hat{\rho}^2\right)/T^{1/2}$; the remaining columns are Newey–West standard errors of Eq. (A.4) at Bartlett bandwidth L, with L^A the adaptive rule of Appendix A.2 (10, 6, 8 for the VIX rows; 5, 4, 4 for r_t^2). Crisis dates are the windows of Table 7 restricted to trading dates.

Table 9. Crisis Discrimination: Area Under the Receiver Operating Characteristic Curve

ID	Type	Days	EPU_t	PU_t	VIX_t	r_t^2	$\hat{\sigma}^2_{\text{BVL},t}$
Panel A. Window by window, against the 954 calm trading dates							
C1	Political	18	0.600	0.947	0.513	0.432	0.219
C2	Political	12	0.373	0.873	0.379	0.531	0.437
C3	Economic	72	0.960	0.668	0.947	0.626	0.928
C4	Political	8	0.098	0.607	0.708	0.740	0.950
C5	Political	76	0.275	0.487	0.423	0.623	0.862
C6	Economic	12	0.806	0.870	0.566	0.430	0.531
C7	Political	8	0.406	0.999	0.648	0.573	0.651
C8	Political	29	0.263	0.915	0.549	0.533	0.542
Panel B. Pooled by episode type							
All windows		234	0.540	0.690	0.629	0.585	0.751
Economic (C3, C6)		84	0.938	0.697	0.893	0.598	0.871
Political (C1–C2, C4–C5, C7–C8)		150	0.318	0.686	0.481	0.578	0.684
Political, excluding C5		74	0.361	0.891	0.540	0.532	0.501

Panel C. Paired difference $\widehat{\text{AUC}}(\text{PU}_t) - \widehat{\text{AUC}}(\text{EPU}_t)$

Subsample	Days	Difference	DeLong z	Block-bootstrap 95% interval
All windows	234	+0.150	5.56	[−0.039, +0.343]
Economic	84	−0.241	−10.17	[−0.361, −0.104]
Political	150	+0.369	14.71	[+0.221, +0.515]
Political, excluding C5	74	+0.530	17.35	[+0.352, +0.681]

Note: entries in Panels A and B are $\widehat{\text{AUC}}$ of Eq. (A.10), the Mann–Whitney estimate of $\Pr(X_t > X_s)$ for a crisis date t and a calm date s, ties weighted one half, on the 1 188 trading dates of Section 4.1; crisis dates from Table 7, dated from the historical record and so exogenous to every series. $\hat{\sigma}^2_{\text{BVL},t}$ is the conditional variance of the GARCH(1, 1) on r_t; because the AUC depends only on ranks, it would be identical for the conditional standard deviation. Panel A widths sum to 235 because C7 and C8 share 20 December 2022. A value below 0.5 is an inverted detector, not a failed one. In Panel C the z uses the full DeLong et al. (1988) covariance (Appendix A.4) but assumes independent days, which the persistence of Table 5 rules out; inference rests on a circular bootstrap of 20-observation blocks (2 000 replications), after pooling dates by crisis status, whose standard errors exceed the DeLong ones by factors of 2.8 to 3.6. The pooled row, significant under DeLong, has a bootstrap interval containing zero.

Table 10. Correlations Among Uncertainty Measures

	EPU_t	VIX_t	r_t^2	$\hat{\sigma}^2_{\mathrm{BVL},t}$
EPU_t	1			
VIX_t	0.4341	1		
r_t^2	0.0497	0.3109	1	
$\hat{\sigma}^2_{\mathrm{BVL},t}$	0.2855	0.6356	0.2600	1
Correlation with EPU_t *by regime, with HAC standard error*				
Full sample	—	0.434 (0.085)	0.050 (0.037)	0.285 (0.093)
Crisis	—	0.648 (0.096)	0.010 (0.066)	0.292 (0.127)
Calm	—	0.261 (0.074)	0.047 (0.042)	0.235 (0.055)

Note: Pearson correlations on the trading-day panel, $T = 1\,188$: EPU_t, the VIX, the squared return r_t^2, and the conditional variance $\hat{\sigma}^2_{\mathrm{BVL},t}$ from the Gaussian GARCH$(1,1)$ of Table 6. All variance measures are in squared decimal units. The Student-t filter delivers a variance correlated at 0.994 with the Gaussian one. Parentheses hold Newey–West standard errors at the adaptive bandwidth.

Table 11. Response of BCRP Foreign-Exchange Intervention to a Political-Uncertainty Shock

Target	Sample	p	T	Impact	95% interval	Granger p
Gross intervention $\lvert\mathrm{INT}_t\rvert$	Full	3	1 183	+10.4	[+0.4, +20.3]	0.016
Gross intervention $\lvert\mathrm{INT}_t\rvert$	Crisis	2	206	+13.2	[+1.6, +27.2]	—
Net spot purchases $\lvert\mathrm{COM}_t\rvert$	Full	3	1 183	+2.4	[+0.5, +4.4]	0.874

Note: impact is the same-day orthogonalized response, in millions of US dollars, to a one-standard-deviation shock (≈ 5.6 index points) in PU_t, from the bivariate VAR of Eq. (14) with the index first in the Cholesky ordering and lag order p by the BIC. Full rows: 1 183 common trading dates, asymptotic intervals. Crisis row: the same system inside the windows of Table 7, stacked as blocks (C4 and C7 too short to enter), percentile intervals from 500 residual-bootstrap replications within window blocks. Lag order is selected separately for each sample. T counts the 1 183 matched dates on the full sample, including the first p dates used as lags, and the 206 observations that enter the regression in the crisis blocks, after the first p dates of each block. On the full sample, the response of $\lvert\mathrm{INT}_t\rvert$ the next day is -12.9, so the cumulative two-day response is -2.6. Granger p tests the null that the coefficients on lags of PU_t are jointly zero; it is not defined for the block design.

C Figures

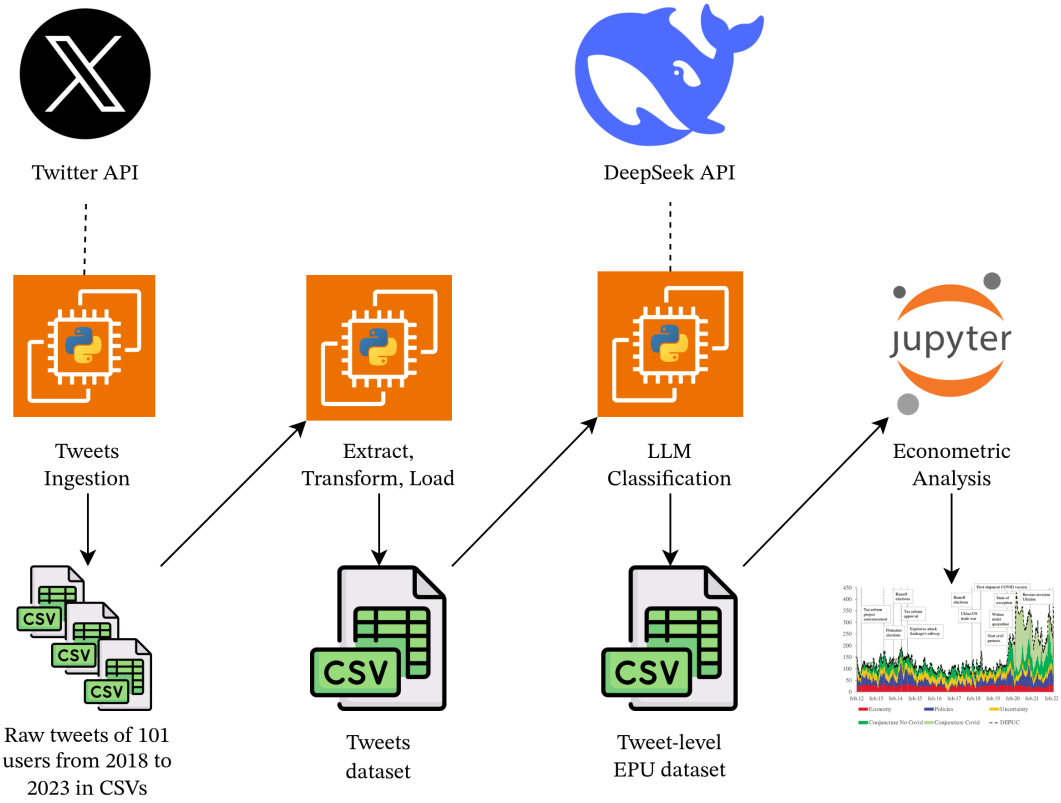


Figure 1. Pipeline Architecture, from Tweet Ingestion to Econometric Analysis

The initial pool comprised 101 accounts; 97 met the selection criteria of Section 2.

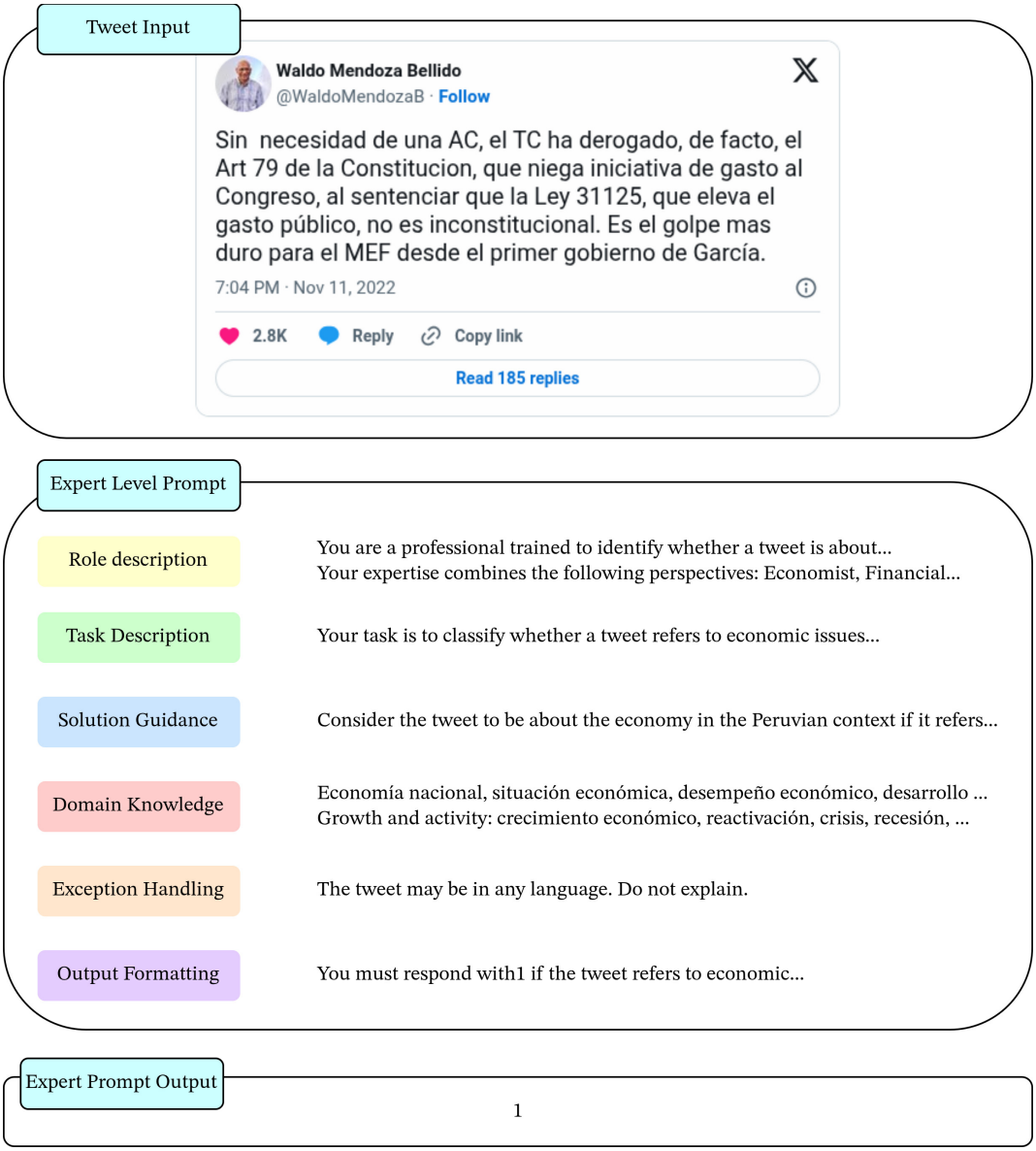


Figure 2. Expert-Level Prompt Structure for LLM Classification

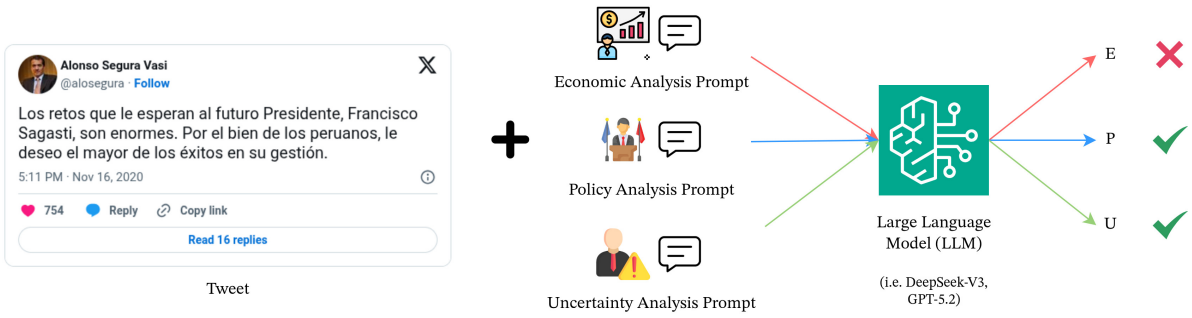


Figure 3. Three-Dimensional Labelling: Each Tweet Is Classified Independently for E_{ijt}, P_{ijt}, U_{ijt}

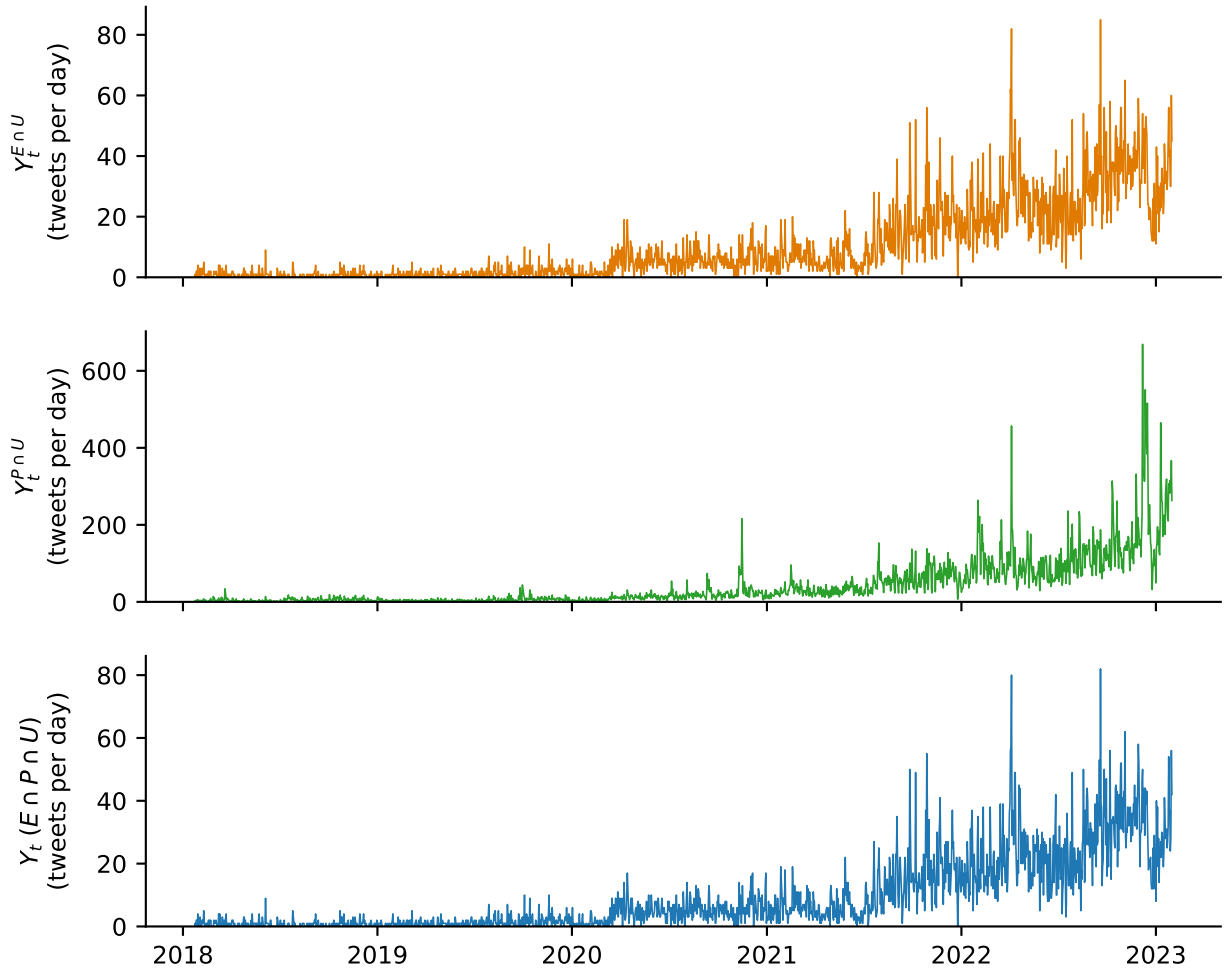


Figure 4. Daily Counts by Cell

Daily counts of tweets that are, top to bottom, economic and uncertain ($Y_t^{E \cap U}$), political and uncertain ($Y_t^{P \cap U}$), and economic, political, and uncertain at once (Y_t, the strict cell of Eq. (1)). The strict count and $Y_t^{E \cap U}$ are nearly identical; $Y_t^{P \cap U}$ is an order of magnitude larger and differently timed. All three rise with posting volume. Sample January 2018–January 2023; 97 accounts.

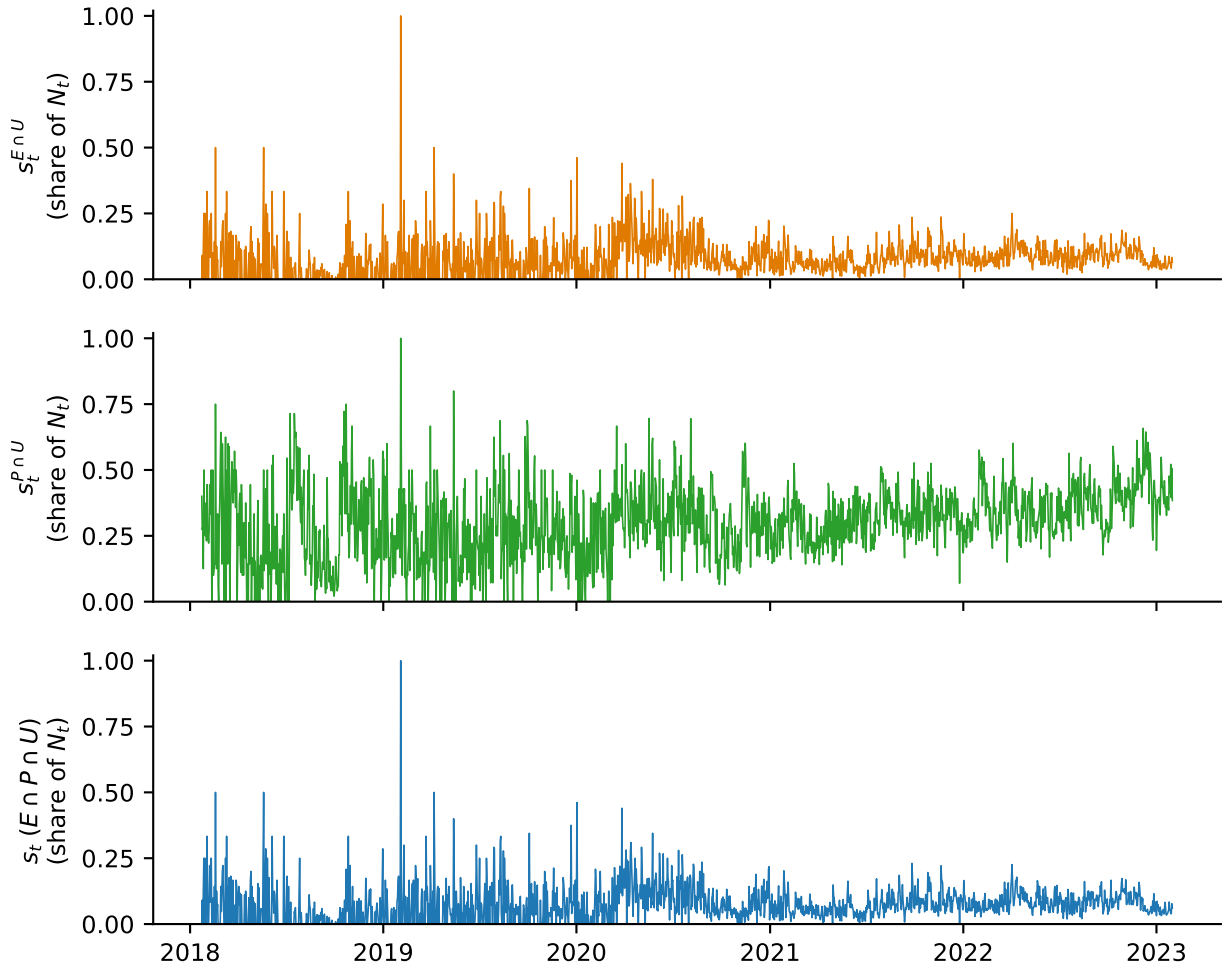


Figure 5. Daily Shares by Cell

Top to bottom, the daily shares of the economic-and-uncertain cell, the political-and-uncertain cell, and the strict cell, each count divided by the daily volume N_t as in Eq. (2). Normalizing by daily volume removes the upward drift of Figure 4 but leaves heavy small-sample noise on low-volume days, when a tiny denominator lets a single tweet move the share sharply.

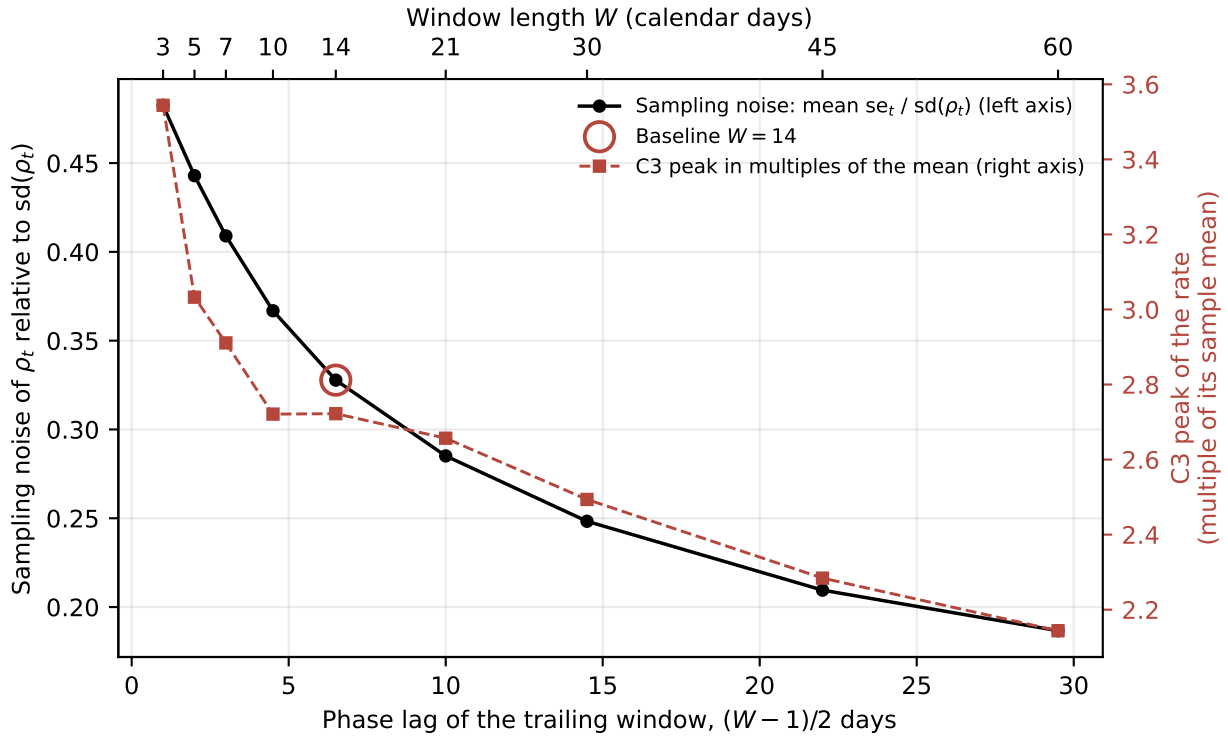


Figure 6. The Window-Length Trade-Off

Each point is one trailing window of W calendar days in Eq. (3), placed at its phase lag $(W-1)/2$ (bottom axis; window length on the top axis). Black, left axis: the sample mean of the binomial standard error se_t of Eq. (5), relative to the standard deviation of ρ_t. Red, right axis: the highest value of ρ_t inside the crisis window C3, in multiples of its sample mean. Both statistics use dates with a complete window, $t \geq W$. The circled baseline $W = 14$ has a 6.5-day lag, a noise ratio of 0.33, and a C3 peak of 2.72 times the mean.

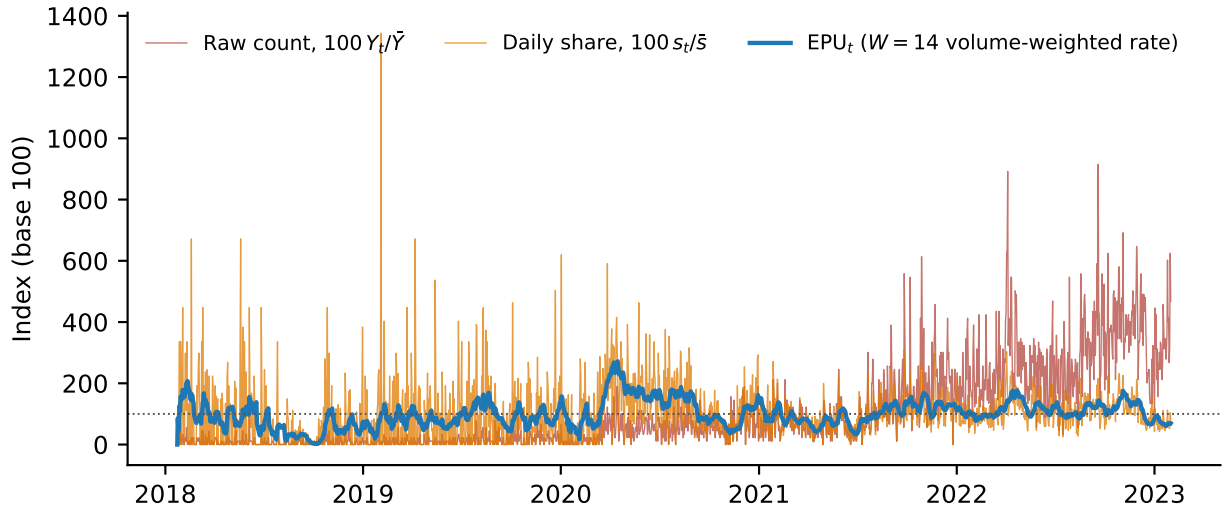


Figure 7. The Index Versus Simpler Forms

EPU$_t$, the 14-day volume-weighted rate of Eq. (3) scaled to base 100 by Eq. (6), against the raw count $100\,Y_t/\bar{Y}$ and the daily share $100\,s_t/\bar{s}$ on the same scale. The simpler forms swing to many times their average; EPU$_t$ stays near its base and retains the episodes. The dotted line marks 100.

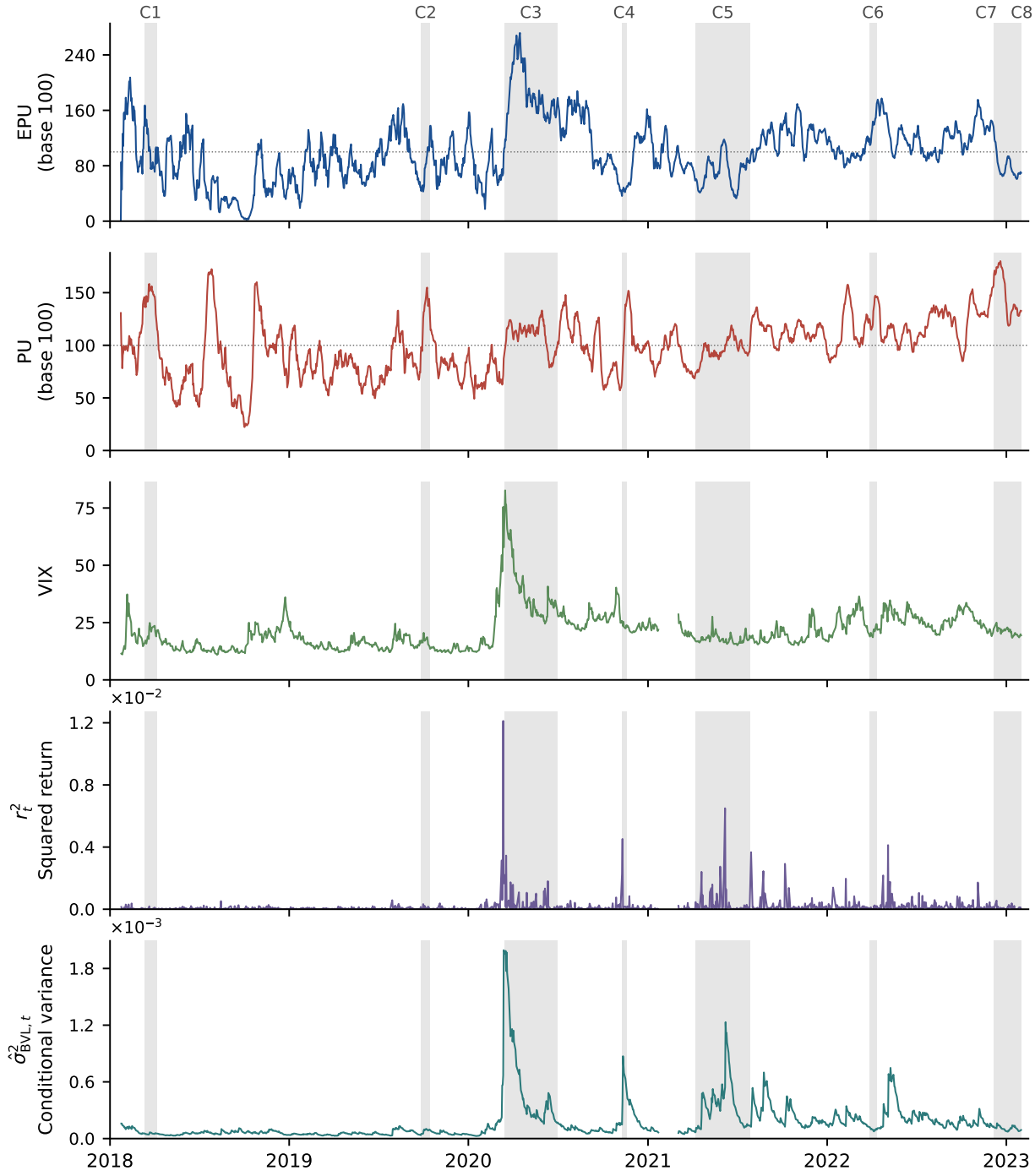


Figure 8. Text Indices and Market Benchmarks Across Crisis Windows

Top to bottom: EPU_t, PU_t, VIX_t, the squared BVL log return r_t^2, and the BVL conditional variance $\hat{\sigma}^2_{\mathrm{BVL},t}$ from the GARCH(1, 1) of Table 6; r_t^2 and $\hat{\sigma}^2_{\mathrm{BVL},t}$ are in squared decimal units. Text indices use all calendar days; market series use the 1 188 trading dates with a valid daily return, and their gap in early 2021 is the excluded BVL stretch. Shaded bands mark C1–C8 (Table 7).

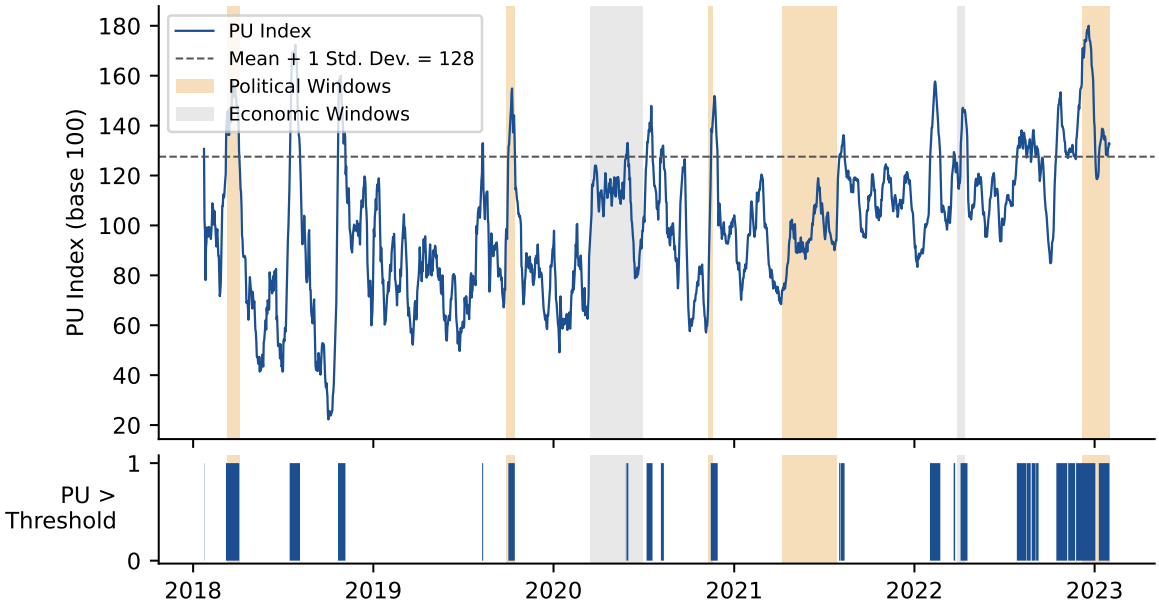


Figure 9. High Political Uncertainty and the Political Crisis Windows

Note: PU_t on all 1 835 calendar days, with political windows in amber, economic windows in grey, and the dashed line one standard deviation above the mean (128). The bottom indicator equals 1 above that threshold and 0 otherwise. Excluding C5, 81% of political-window days lie above it, against 13% of calm days; no C5 day does, the weakness Table 9 records. The sample maximum (180) falls on 20 December 2022, where C7 meets C8.

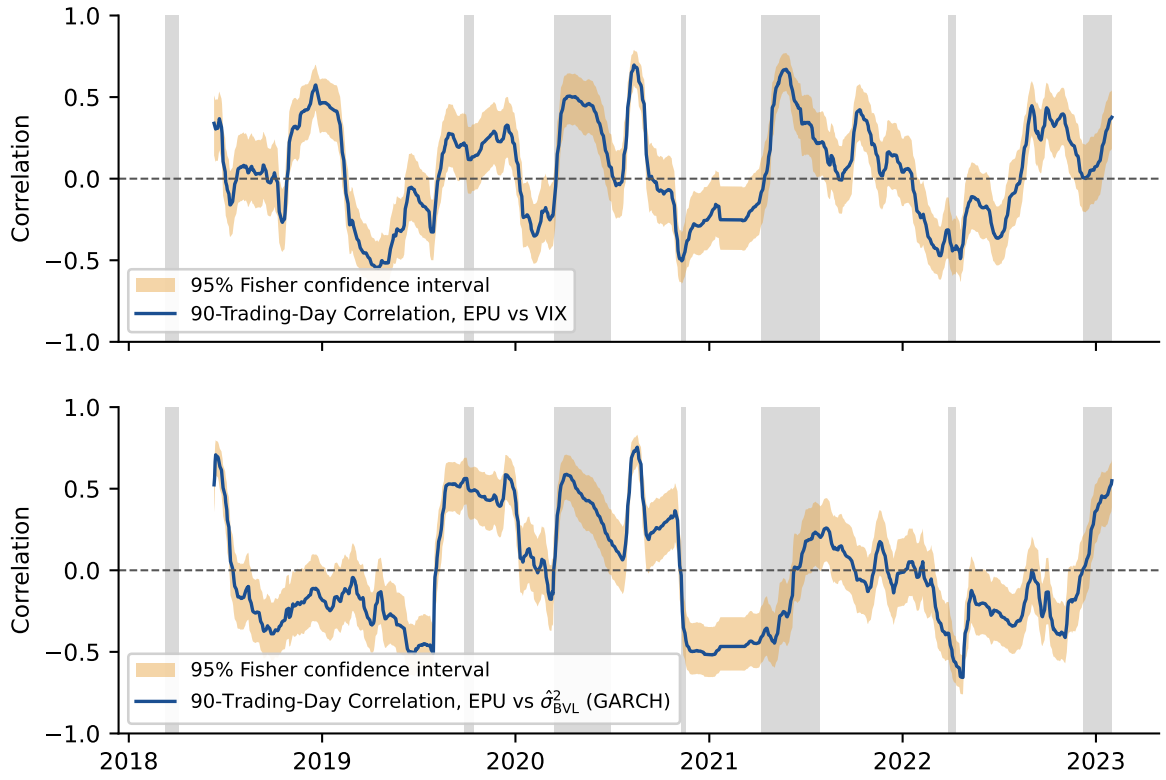


Figure 10. Rolling Correlation of EPU_t over 90 Trading Dates with the VIX and with the BVL Conditional Variance

Note: rolling Pearson correlation of Eq. (10) between EPU_t and b_t, computed over the $m = 90$ trading dates that end at t and centred on the window means $\overline{\text{EPU}}_t$ and $\bar{b}_t$; top, $b_t = \text{VIX}_t$; bottom, $b_t = \hat{\sigma}^2_{\text{BVL},t}$, the GARCH conditional variance of the Lima return. The solid line is the correlation, the shaded band its 95% Fisher reference interval, the dashed rule zero, and the grey bands the crisis windows of Table 7. The Fisher band assumes independence and is descriptive under the serial dependence present here.

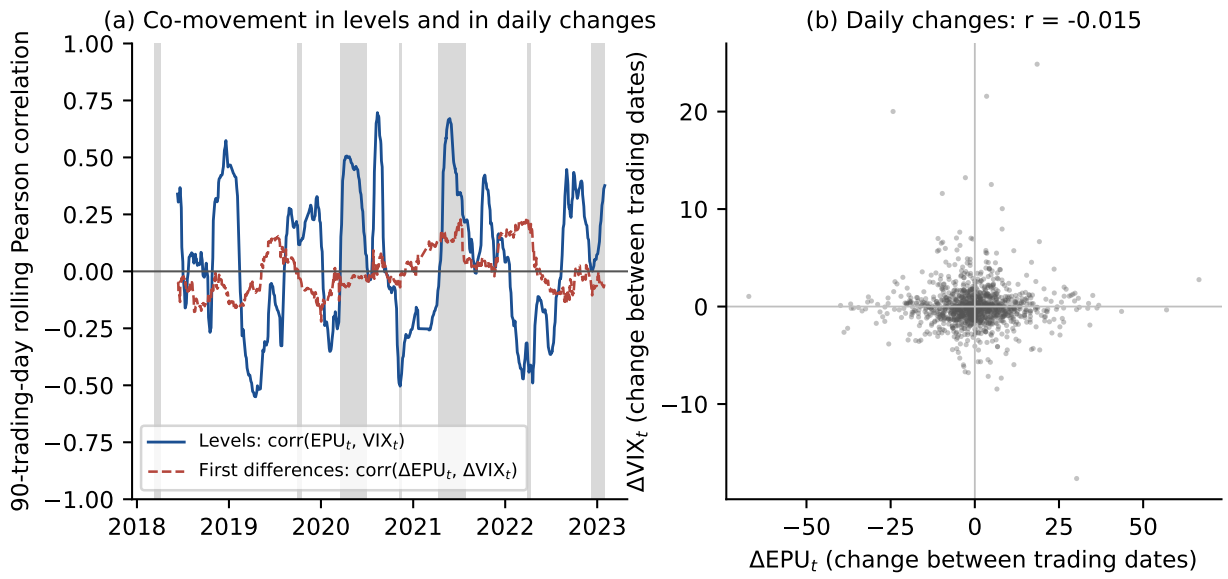


Figure 11. Co-Movement of EPU$_t$ and the VIX in Levels and in First Differences

Note: (a) rolling Pearson correlation of Eq. (10) over 90 trading dates between EPU$_t$ and VIX$_t$ in levels (solid) and between ΔEPU$_t$ and ΔVIX$_t$ (dashed), with the crisis windows C1–C8 shaded; (b) scatter of ΔEPU$_t$ against ΔVIX$_t$. Differences are taken between consecutive dates of the trading-day panel. The level correlation swings with the crisis regimes, while the difference correlation stays near zero throughout (-0.015 over the sample), so the similarity is a low-frequency, between-episode agreement rather than a daily one.

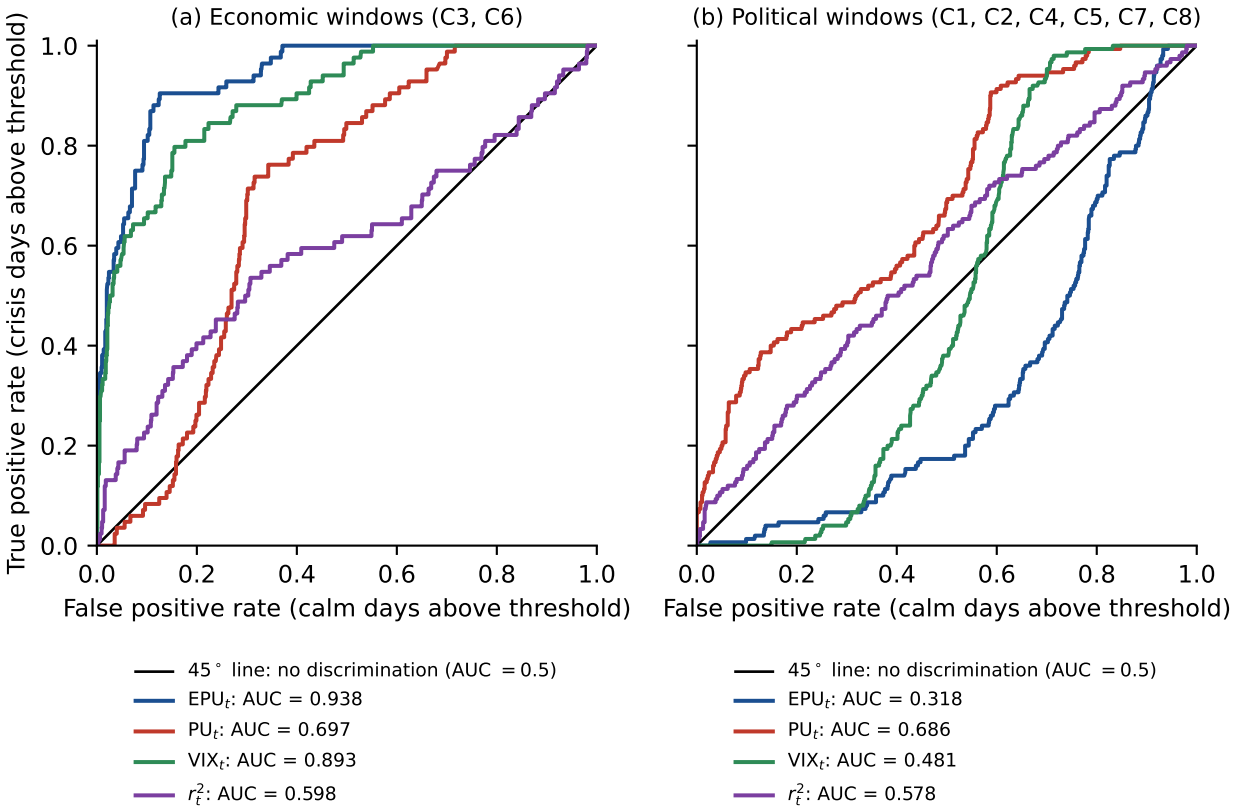


Figure 12. Receiver Operating Characteristic Curves for Crisis Days

Note: panel (a) pools the economic windows C3 and C6 (84 trading dates), panel (b) the six political ones (150); the negative class is the same 954 calm dates, so the curves reproduce Table 9. Each curve traces the true positive rate (crisis dates above a threshold) against the false positive rate (calm dates above it) as the threshold moves. The solid black line is the 45° line, on which the two rates are equal and the AUC is 0.5; a curve above it ranks crisis dates higher than calm dates, and a curve below it is an inverted detector. Legend entries report $\widehat{\text{AUC}}$.

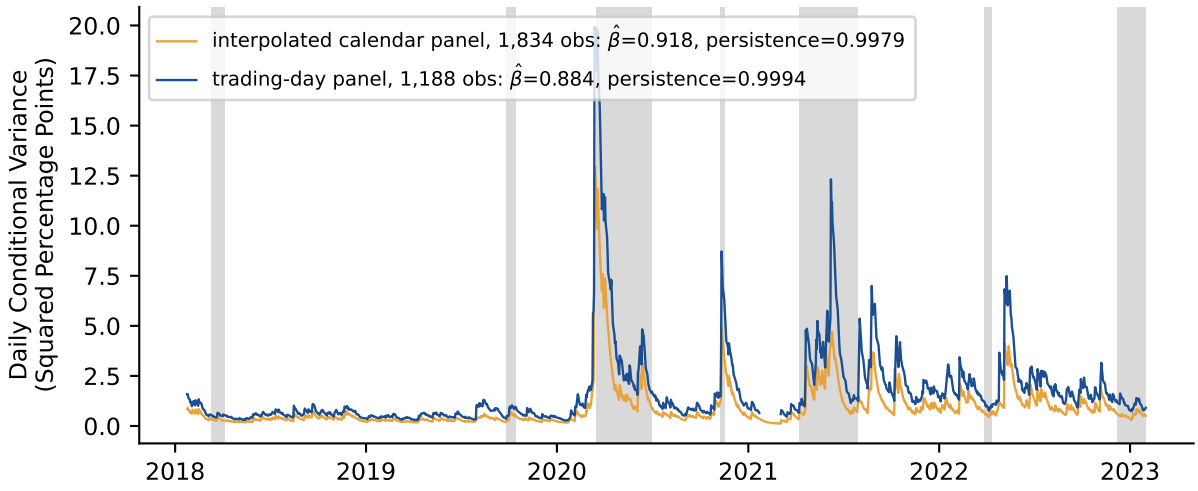


Figure 13. Effect of Interpolating Non-Trading Dates on the Conditional Variance

Note: conditional variance of the BVL log return from the same Gaussian GARCH$(1,1)$ estimated twice, once on the interpolated calendar panel of 1 834 returns and once on the 1 188 valid returns of the trading-day panel, in squared percentage points. Interpolation lowers the measured variance and changes the fitted parameters (Table 6). The trading-day line is broken over the excluded BVL stretch of early 2021.

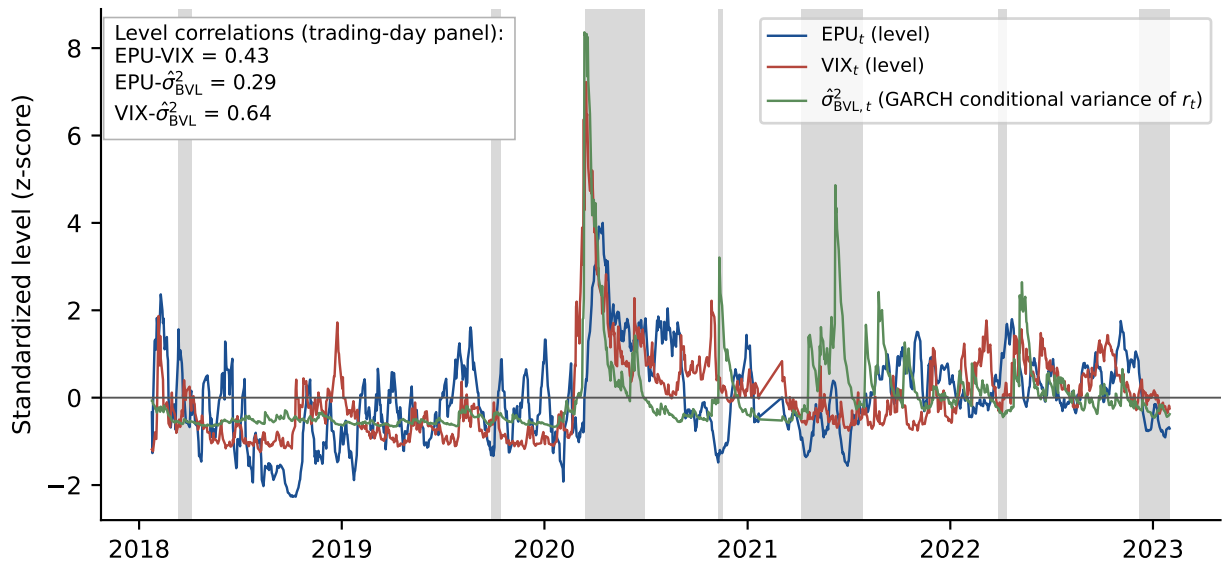


Figure 14. Three Levels of Uncertainty on the Same Plane

Standardized levels of EPU$_t$, VIX$_t$, and $\hat{\sigma}^2_{\text{BVL},t}$, the conditional variance of BVL returns from Table 6, on the trading-day panel, with C1–C8 shaded. The three rise sharply during C3; their responses differ across the other windows. Pairwise correlations are in Table 10.

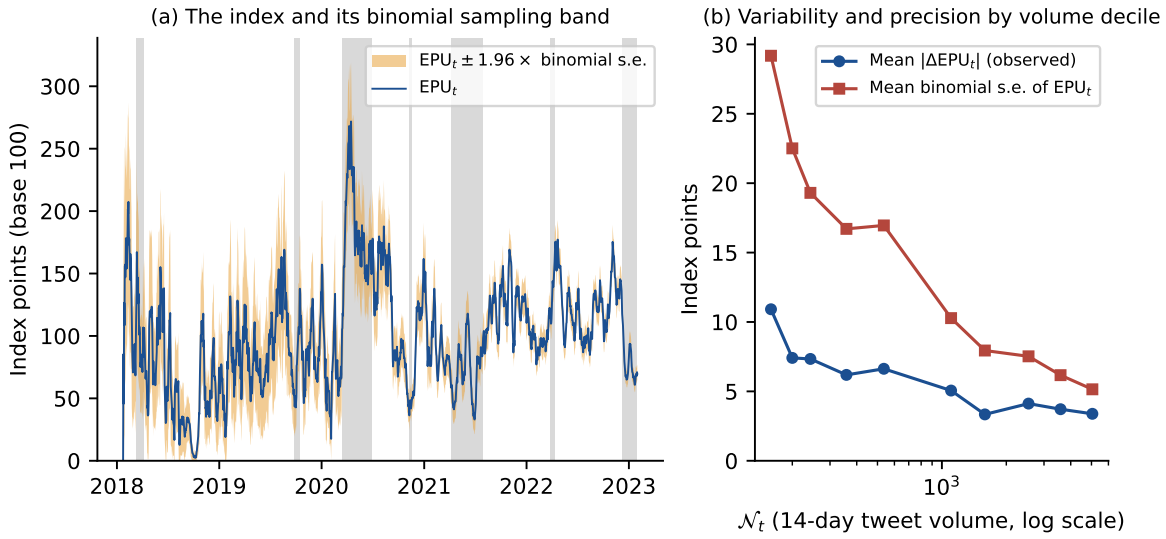


Figure 15. Tweet Volume and the Sampling Precision of EPU$_t$

(a) EPU$_t$ on all calendar days with a band of ±1.96 binomial standard errors, $100\,\mathrm{se}_t/\bar{\rho}$ with se_t from Eq. (5); the band is widest in 2018–2019, when few tweets enter each window. (b) For deciles of $\mathcal{N}_t$, the number of tweets in the fourteen calendar days ending on day t, the mean absolute daily change $|\Delta\mathrm{EPU}_t|$ (observed variability) and the mean binomial standard error, both in index points, against the median $\mathcal{N}_t$ of each decile (log scale). Both fall with volume; the association does not identify the share of variability due to measurement error. No GARCH model is fitted to the index.

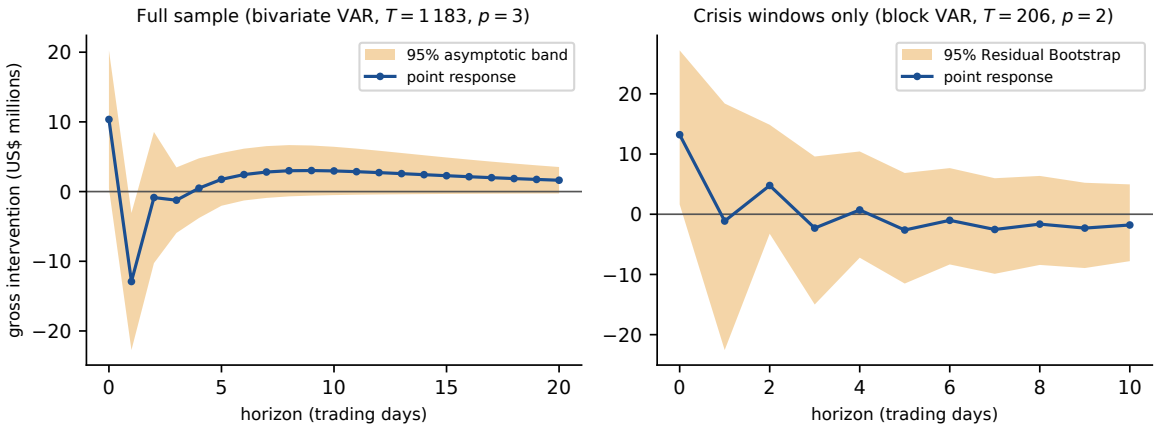


Figure 16. Response of BCRP Gross Foreign-Exchange Intervention to a Political-Uncertainty Shock

Note: orthogonalized responses of gross intervention $|\mathrm{INT}_t|$, in millions of US dollars, to a one-standard-deviation PU_t shock (≈ 5.6 points). Left: VAR on 1 183 trading dates, $p = 3$, asymptotic 95% bands. Right: the same system inside the crisis windows only, $p = 2$, stacked as blocks (C4 and C7 too short to enter), with residual-bootstrap bands. Lag order is selected separately by sample. Impact +10.4 million on the full sample, reversed the next day (-12.9), and +13.2 inside the windows; Table 11 collects the estimates.

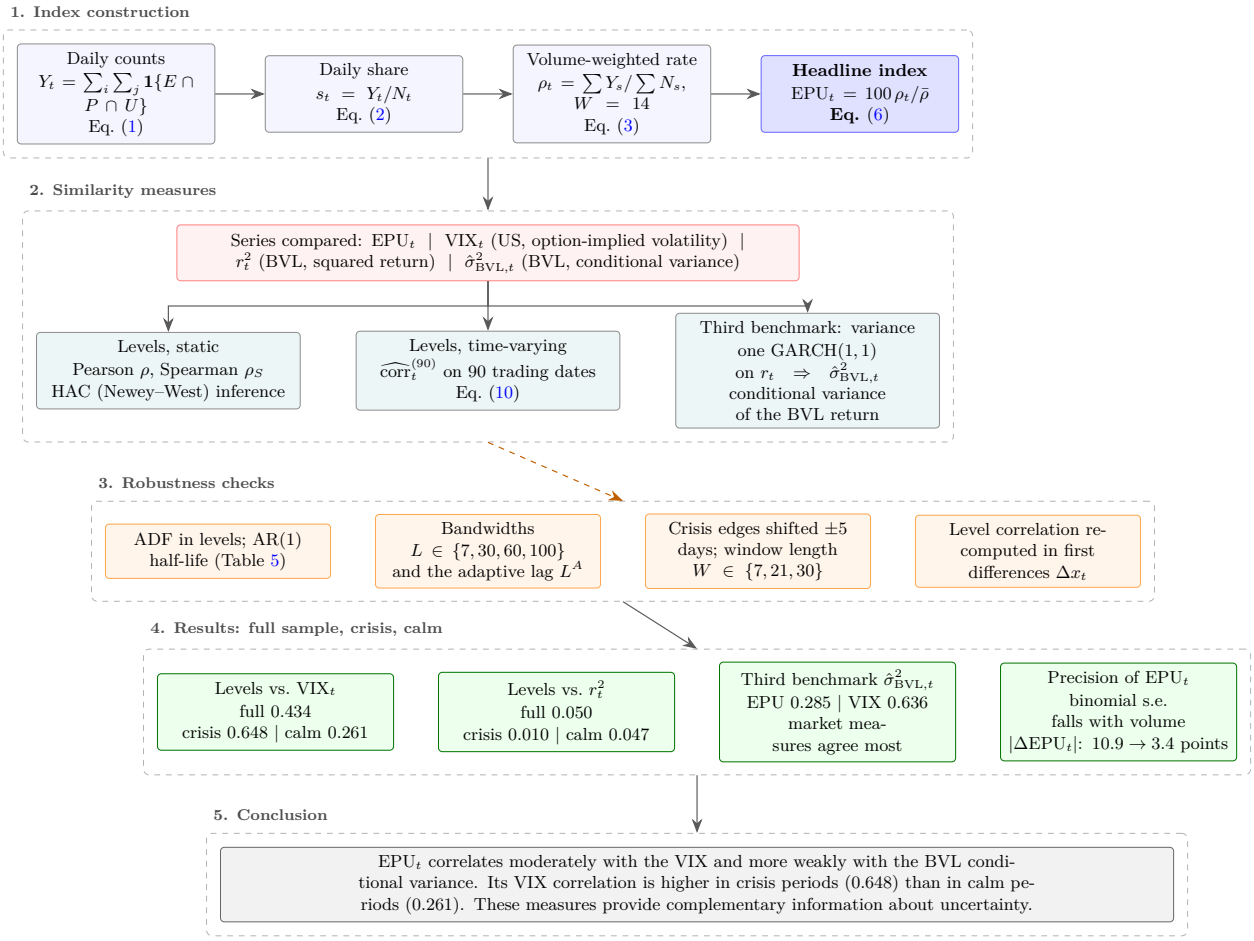


Figure 17. Synthesis of the Paper

The construction of the headline index (top), the measures of similarity applied to the benchmarks (middle), the diagnostics that discipline them, the results on the full sample and on the crisis and calm subsamples (bottom), and what the evidence licenses (foot). Solid arrows carry data; dashed arrows carry robustness checks. The numbers in the results row are Pearson correlations computed on the trading-day panel of Section 4.1, except the changes in index points of the precision box; the other numbers are design parameters (the 14-day window, the alternative windows, the 90-date rolling window, the ±5-day shifts, and the HAC bandwidths).

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